DIGI-RATION

A MINI - PROJECT REPORT

Submitted by

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

MEPCO SCHLENK ENGINEERING COLLEGE, SIVAKASI

(An Autonomous Institution affiliated to Anna University Chennai)

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***Register No.* :** ……………………….

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This is to certify that it is the Bonafide work titled **"DIGI-RATION"** carried out by MALESHKUMAR V (Reg. No.9517202304076**)**, SUJAN S (Reg. No.9517202304164), PANDI S (Reg. No.9517202304109) for the **23CS453 - Mini Project – II**, Mepco Schlenk Engineering College (Autonomous), Sivakasi during **IV semester** in the academic year 2024 – 2025.

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…………………………. …………………………

Internal Examiner - I **Internal Examiner - II**

ABSTRACT

The development of a web-based ration distribution system aims to address the inefficiencies prevalent in traditional ration shops by automating the entire process through PHP and MySQL technologies. By eliminating manual procedures, the system facilitates a more streamlined distribution of rations, ensuring fair allocation and significantly reducing wait times for beneficiaries. Key features such as virtual token generation, real-time stock tracking, and digital ration card management are designed to enhance user accessibility while simultaneously improving government oversight and monitoring of ration distribution practices.

In addition to improving efficiency and accessibility, the system also tackles issues of inconsistent data. A transparent complaint resolution system will further empower users to report issues without fear of retribution, fostering greater trust in the system. By integrating these innovative solutions, the web-based ration distribution system not only enhances the overall experience for consumers but also provides authorities with critical tools to oversee stock movements and ensure the integrity of the distribution process, ultimately leading to a fairer and more transparent rationing framework.

ACKNOWLEDGEMENT

First and foremost, we thank the **LORD ALMIGHTY** for his abundant blessings that is showered upon our past, present and future successful endeavors.

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LIST OF ABBREVIATIONS

ABBREVIATIONS	DESCRIPTION
API	Application Programming Interface
PDS	Public Distribution System
PHP	Hypertext Preprocessor
HTML	Hypertext Markup Language
SQL	Structured Query Language
JSON	JavaScript Object Notation
HTTP	Hypertext Transfer Protocol
TPDS	Targeted Public Distribution System
EPDS	Electronic Public Distribution System

CHAPTER 1

INTRODUCTION

The Public Distribution System in India (PDS) is not only for a group of users and it is also for the whole country. PDS plays an important role in the development of a country. In PDS system people can directly get their food products in low price PDS also plays a great role in maintaining the identity of the people all over the country.

1.1 PROBLEM STATEMENT

The existing ration distribution system suffers from a severe lack of transparency in stock allocation, leading to mismanagement, corruption, and unequal distribution of essential goods. Admins and distributors are often unable to track inventory in real time, resulting in inefficiencies. Moreover, the manual process of onboarding and verifying distributors using physical documents causes delays and errors, demanding a secure and digitized registration process that includes Aadhar, Ration Card, and PDS number validation. There is also no reliable way to track inactive or non-compliant distributors, which creates loopholes in monitoring and accountability. The absence of a structured system to track stock requests by distributors leads to ration delays, highlighting the need for a proper 'Apply for Stock' and 'View Applied Stock' mechanism. From the customer's perspective, unawareness of product availability results in unnecessary trips to ration shops and dissatisfaction; a real-time product dashboard could resolve this issue effectively.

In addition, the limited payment options currently available restrict customers to manual transactions, underlining the need for integrated offline and online payment modes during ration booking. Both distributors and customers also face the issue of no access to digital transaction records or statements, making it hard to verify transactions or maintain history. The lack of an effective complaint redressal mechanism prevents customers from voicing their grievances directly to the authority, reducing service quality. Furthermore, physical ration cards are outdated and prone to loss or damage, which can be resolved by introducing a digital e-ration card system with family and distributor details. Lastly, the absence of personalization and accessibility features such as profile editing and dark mode negatively impacts user experience, especially for users with different needs or preferences. These challenges clearly demonstrate the necessity for a digitized, role-based, and user-friendly platform that streamlines ration management while ensuring transparency, accountability, and accessibility.

1.2 OBJECTIVES

The primary objective of this system is to digitize and streamline the ration distribution process by offering a centralized platform that facilitates efficient coordination between the admin, distributor, and customer. It aims to ensure transparency in stock allocation and distribution, enabling the admin to monitor real-time data related to customers, distributors, complaints, and stock requests through an interactive dashboard. A key objective is to simplify the distributor registration process by collecting and validating essential details such as Aadhar, Ration Card, PDS number, and shop address through a secure and user-friendly interface. The system also seeks to provide the admin with full control to add stocks, assign rations, and suspend inactive distributors, thereby improving accountability and minimizing manual overhead.

Another important goal is to empower distributors and customers with access to real-time data. Distributors can apply for stock when there is a shortage, track transactions, view statements, and manage customer relationships, while customers can browse available products, book rations, make payments (offline or online), and access their e-ration cards. The system intends to facilitate a smooth complaint submission process, allowing customers to directly raise issues to the admin, enhancing the overall service quality. Additionally, it strives to offer a personalized and accessible experience by enabling profile management and theme customization (like dark mode). Overall, the system is designed to eliminate inefficiencies, reduce corruption, and ensure fair and timely delivery of rations to the beneficiaries.

1.3 SCOPE

The scope of this Ration Distribution Management System extends across the full spectrum of digital public distribution management, focusing on enhancing the efficiency, transparency, and user experience for administrators, distributors, and customers. The system is designed to be implemented at local, regional, or even national levels where rationing services are provided, making it adaptable for various public welfare departments and government schemes. It allows the admin to manage and monitor all distribution activities from a centralized dashboard, including adding new distributors, assigning stock based on PDS numbers, viewing stock requests, tracking complaints, and monitoring distributor activity for suspension if necessary. The scope also includes secure distributor onboarding with complete verification, stock management features, and the ability to track every product movement within the distribution chain.

For distributors, the system covers functionalities like applying for stock, monitoring stock levels, viewing customer lists, tracking transactions, and editing their profiles, thereby enabling them to efficiently manage their operations. From the customer's perspective, the scope includes features to view available products, book rations, make payments, track transaction history, access their digital ration card, raise complaints, and customize their user interface with options like dark mode. The system is built with scalability in mind, allowing for future integration with biometric authentication (e.g., Aadhaar), SMS alerts, multilingual support, and mobile applications. Ultimately, the scope encompasses the creation of a robust, secure, and accessible ecosystem that replaces the outdated manual processes of ration distribution and contributes to a more accountable and citizen-friendly welfare service.

CHAPTER 2

EXISTING AND PROPOSED MODEL

2.1 EXISTING MODEL

The functional requirements in the existing Ration Distribution System highlight the limited operational capabilities and inefficiencies in managing ration distribution among customers, distributors, and admins.

1. It provides basic functionality for the admin to manage distributors, but relies on static dashboards with manual record-keeping, resulting in delays and inefficiencies.
2. Admins can add distributors using paper forms, leading to potential errors and inefficiencies in capturing distributor data.
3. Stock management is primarily manual, making it challenging to track stock levels and allocate supplies appropriately, often leading to shortages or surplus.
4. Distributors and customers face cumbersome processes, including manual tracking of transactions and reliance on physical ration cards, which increases the risk of loss or incorrect information.
5. The absence of a centralized complaint management system makes it difficult for customers to raise grievances and for admins to respond in a timely manner.
6. Customers have no way to check available products in real-time, often leading to unnecessary travel and miscommunication.
7. Distributor stock requests are usually handled offline, lacking transparency and proper documentation.
8. There is no provision for online or digital payments, making the booking process limited to physical transactions.
9. Physical ration cards lack digital backup or retrieval, risking permanent data loss if damaged or misplaced.
10. The interface lacks personalization and accessibility options such as profile editing, visual themes, or adaptive modes.

2.2 PROPOSED MODEL

The proposed Ration Distribution Management System addresses the deficiencies in the existing model through enhanced functionalities and improved user experience.

1. The system will feature a dynamic admin dashboard with real-time metrics, allowing admins to monitor essential statistics such as total customers, complaints, distributors, and requested stocks effectively.
2. Distributor onboarding will be facilitated through a streamlined online form that collects comprehensive data, ensuring accuracy and efficiency in the process.
3. Automated stock management will allow admins to add stock based on PDS numbers and assign it easily to all distributors, thereby streamlining the distribution process.
4. The system will offer a digital interface for distributors to manage customer interactions, submit stock requests, and view transactions, replacing manual records with automated tracking.
5. Customers will benefit from an intuitive dashboard that enables easy access to ration booking, transaction history, and e-ration cards, thus simplifying their experience and improving accessibility.
6. A built-in complaint submission and tracking feature will empower customers to voice concerns, while admins can address them promptly within the system.
7. The real-time product visibility allows customers to view available stock before booking, minimizing unnecessary visits or misunderstandings.
8. Online and offline payment support will provide flexibility to users, improving convenience and modernizing the ration booking process.
9. Digital e-ration cards will be available with family details and linked distributor information, ensuring secure access anytime.
10. The platform will support user profile customization and accessibility features like dark mode, offering a user-friendly and inclusive experience.

CHAPTER 3

SYSTEM DESIGN

3.1 SOFTWARE REQUIREMENTS SPECIFICATION / COMPONENTS REQUIREMENTS SPECIFICATION

Introduction

Purpose of the Requirements Document

This Software Requirements Specification (SRS) document describes the complete specifications for the Ration Distribution Management System (RDMS), designed to streamline the distribution of rations to customers, facilitate smooth operations for administrators, and ensure efficient management for distributors. The RDMS will replace manual processes with automated workflows, improving transparency, security, and efficiency in ration distribution. The purpose of this document is to define the system's requirements, providing a blueprint for developers, designers, and stakeholders. This document is intended to serve as a basis for development and further refinement, offering an in-depth overview of functionalities, user interactions, and system constraints.

Scope of the Product

The Ration Distribution Management System is aimed at digitizing the entire process of ration allocation, from stock management to customer delivery. The product will support three primary user groups: administrators, distributors, and customers. The system will allow administrators to manage stock, distributor details, and customer complaints. Distributors will be able to track stock levels, request stock from admins, and view customer information. Customers will have the ability to book rations, make transactions, and track their history through an e-ration card. The scope of this system includes providing real-time updates, transaction tracking, complaint resolution, and ensuring compliance with government regulations related to public distribution.

Definitions, Acronyms, and Abbreviations

1. Admin: A user role responsible for overseeing the overall ration distribution process, managing stock, users, and complaints.
2. Distributor: An intermediary responsible for distributing rations to customers based on the stock allocated by the admin.
3. Customer: The end-user of the system who receives rations from distributors.
4. e-Ration Card: A digital version of the ration card used by customers to book rations.
5. PDS: Public Distribution System, the government system for ration distribution.
6. Stock Allocation: The process of assigning rations to distributors based on their requests.
7. Complaint Resolution: The handling of complaints submitted by customers regarding their ration services.

References

1. Public Distribution System regulations as set by the Government of India.
2. Indian Government's Digital Infrastructure Guidelines.
3. ISO/IEC 27001:2013 standards for information security management.
4. Relevant online payment systems and integration APIs (e.g., Paytm, UPI).

Overview of the Remainder of the Document

This document is structured into several sections. Section 2 provides a detailed general description of the product, including its user characteristics and interfaces. Section 3 delves into the specific requirements, including functional and non-functional specifications, and defines the design constraints. The appendices will include diagrams, flowcharts, and additional technical information required for the development phase.

General Description

Product Perspective

The Ration Distribution Management System (RDMS) is a cloud-based solution designed to replace traditional ration distribution systems. It provides a unified platform where administrators can oversee stock management, distributors can manage ration allocation, and customers can book rations. The system's architecture is modular, allowing for ease of maintenance and future scalability.

a. System Interfaces

The RDMS will interface with third-party systems, particularly government databases for Aadhar and PDS verification. This ensures that only authorized individuals can request and receive rations. Additionally, the system will support payment gateway integration for online transactions.

b. User Interfaces

The RDMS will have separate user interfaces for admins, distributors, and customers. Each interface will be designed according to the role-specific needs:

1. Admin Interface: A comprehensive dashboard providing insights into stock levels, complaints, and distributor performance.
2. Distributor Interface: A user-friendly dashboard that allows distributors to view stock availability, apply for stock, and manage customer relations.
3. Customer Interface: An intuitive interface for customers to book rations, view transaction history, and track complaints.

c. Hardware Interfaces

The RDMS is a web-based platform that will work on desktops, laptops, tablets, and smartphones. The system does not require specialized hardware and can operate on common devices equipped with internet access.

d. Software Interfaces

The system will integrate with external software such as payment gateways (Paytm, UPI) and government verification systems. These integrations ensure secure and smooth transactions, as well as the validation of customer identities.

e. Communication Interfaces

The RDMS will utilize secure HTTP/HTTPS protocols for communication between the client (user) and server. Additionally, email and SMS notification services will be integrated to alert users about transaction statuses, stock availability, and complaint resolutions.

f. Memory Constraints

Given the cloud-based nature of the system, memory constraints are not an issue for end-users. On the server-side, the system will employ scalable cloud infrastructure, ensuring adequate memory resources for all users.

g. Operations

The system will support round-the-clock operations. This includes real-time updates on stock availability, instant transaction logging, and immediate complaint handling. The system will operate in a high-availability environment, ensuring minimal downtime.

h. Site Adaptation Requirements

The system will support multiple regional languages and local customizations to cater to the diverse needs of customers across India. It will also comply with local government regulations and infrastructure guidelines.

Product Functions

The RDMS will offer the following key functions:

1. Admin Functions: Manage distributors, stock, complaints, and customer data; allocate stock based on distributor requests; track complaints and transaction history.
2. Distributor Functions: View available stock, apply for stock, manage customer relations, and track transaction statements.
3. Customer Functions: View available products, book rations, view e-ration card, and track transactions. Customers can also submit complaints and edit their profile.

User Characteristics

1. Admin: Requires advanced data access and control, with responsibilities ranging from stock management to distributor and complaint oversight.
2. Distributor: Requires access to stock details, transaction records, and customer management features.
3. Customer: Requires simple access to ration booking, transaction history, and complaint submission.

General Constraints

1 The system must comply with all relevant data protection and privacy laws, including the Indian IT Act and guidelines set forth by the Government of India.

2 The system must be operable on both Android and iOS devices.

Assumptions and Dependencies

3 The system assumes that all users have internet access and can access the platform via a web browser or mobile app.

4 The system will depend on the availability of government databases for identity verification and the stock allocation process.

Specific Requirements

External Interface Requirements

1. Payment Gateway Integration: The system must integrate with Paytm, UPI, and other popular Indian payment systems to handle both online and offline payments.
2. Government Database Integration: The system must interface with government verification systems, such as Aadhar and PDS, to verify customer eligibility.

Functional Requirements

1. Admin Functions: The admin can add new distributors, assign stock, manage complaints, and oversee transaction records. The admin interface will allow suspending non-performing distributors.
2. Distributor Functions: Distributors can check available stock, apply for stock, manage customers, and track transactions.
3. Customer Functions: Customers can book rations, make payments, track their e-ration card, and submit complaints.

Performance Requirements

1. The system should support up to 10,000 concurrent users at peak times.
2. Real-time stock updates must occur within 2 seconds of a change.
3. Transaction data must be updated and recorded immediately to ensure transparency.

Design Constraints (Standard Compliance)

1. The system must adhere to the ISO/IEC 27001 standard for information security.
2. It must be compatible with Android, iOS, and web platforms, ensuring maximum accessibility.
3. The system should support multi-language support, including regional languages.

Logical Database Requirements

1. The system will use a Relational Database Management System (RDBMS), with separate tables for users, stock, transactions, complaints, and payments.
2. The database must support full transaction history and provide logs of user actions for auditing.

Software System Attributes

a. Reliability

1. The system will be reliable, offering 99.9% uptime and regular backups to prevent data loss.

b. Security

1. All user data will be encrypted using SSL/TLS protocols. Two-factor authentication will be implemented for admin and distributor accounts to enhance security.

c. Portability

1. The system will be accessible on any device with an internet connection. It will be designed for cross-platform compatibility, including Android and iOS mobile apps.

d. Availability

1 The system will be operational 24/7, with scheduled downtime for maintenance once a week.

e. Maintainability

2 The system will be modular, with well-documented code to allow easy maintenance and updates.

f**. Other Requirements**

1. The system will include a user-friendly interface, minimal latency, and multilingual support to cater to a wide range of users.

Appendices

Appendix A: Glossary

- Admin: A user role responsible for overseeing ration distribution, managing stock, distributors, and complaints.
- Distributor: An intermediary responsible for distributing rations to customers based on the allocation from the admin.
- Customer: The end-user of the system who receives rations from distributors.
- e-Ration Card: A digital version of the ration card used by customers for ration booking and verification.
- PDS: Public Distribution System, a government system for ration distribution to eligible citizens.
- Stock Allocation: The process of assigning available stock to distributors based on their requests.
- Complaint Resolution: The handling and processing of complaints submitted by customers regarding ration distribution.
- Payment Gateway Integration: The integration of third-party services such as Paytm or UPI for online payment transactions.
- Multi-Language Support: The system's ability to cater to users speaking different regional languages.
- Real-Time Updates: Immediate changes reflected in the system, such as stock updates and transaction logging.
- Two-Factor Authentication (2FA): A security process requiring two forms of identification to access an account.
- Relational Database Management System (RDBMS): A database system that uses structured query language (SQL) to manage and store data.
- SSL/TLS Encryption: Security protocols used to encrypt data transmitted between the server and the user to ensure privacy.

- Cross-Platform Compatibility: The ability for the system to run on multiple platforms, such as Android, iOS, and web browsers.
- ISO/IEC 27001: An international standard for information security management systems, ensuring secure processing and storage of data.

3.2 UML DIAGRAMS

3.2.1 USECASE DIAGRAM

The use case diagram depicts the system's functionalities and interactions between the Admin, Distributor, and Customer. It outlines the actions each role can perform, such as adding distributors, managing stock, processing transactions, and booking rations, within the ration distribution system.

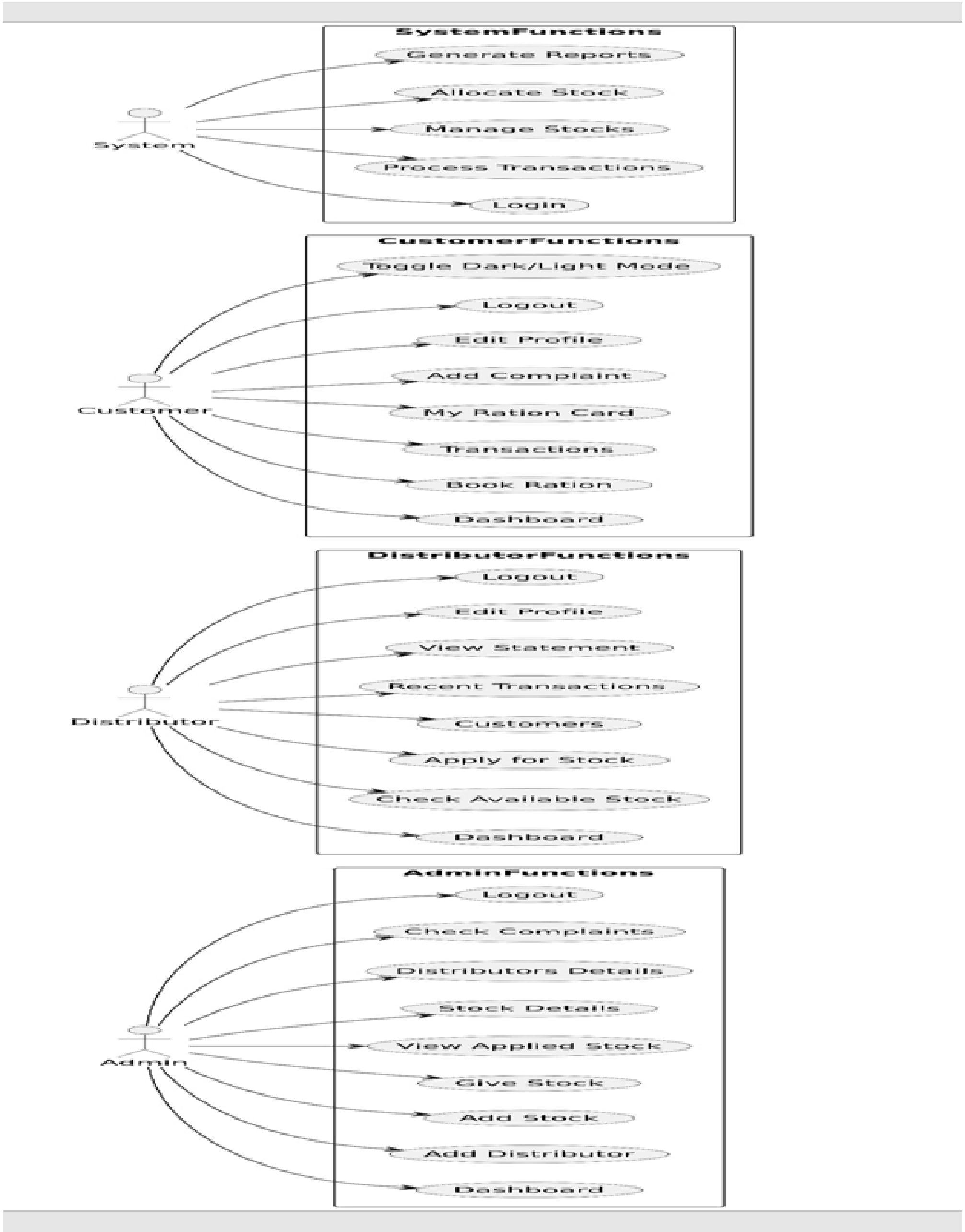

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Figure 3.2.1: Usecase Diagram for Digiration

3.2.2 CLASS DIAGRAM

The class diagram defines the system's structure by representing key entities like Admin, Distributor, Customer, Stock, and Complaint along with their attributes and relationships. It outlines how these classes interact to support functionalities like stock management, user registration, transaction tracking, and complaint handling

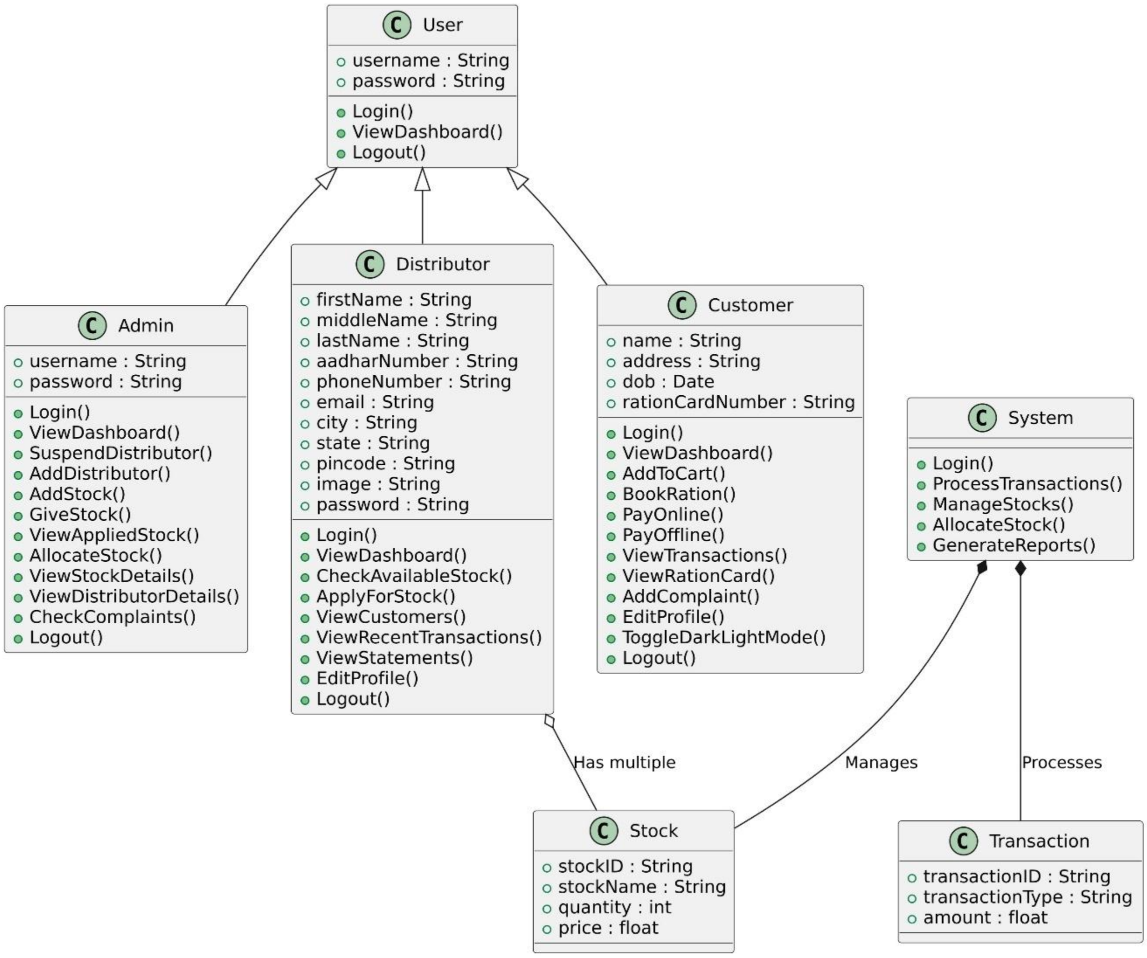

Figure 3.2.2: Class Diagram for Digiration

3.2.3 ACTIVITY DIAGRAM

The system manages ration distribution through three roles: Admin, Distributor, and Customer. The Admin oversees stock allocation, distributor management, and complaints, while the Distributor handles stock requests, customer interactions, and transactions, and the Customer books rations, views transactions, and manages their profile.

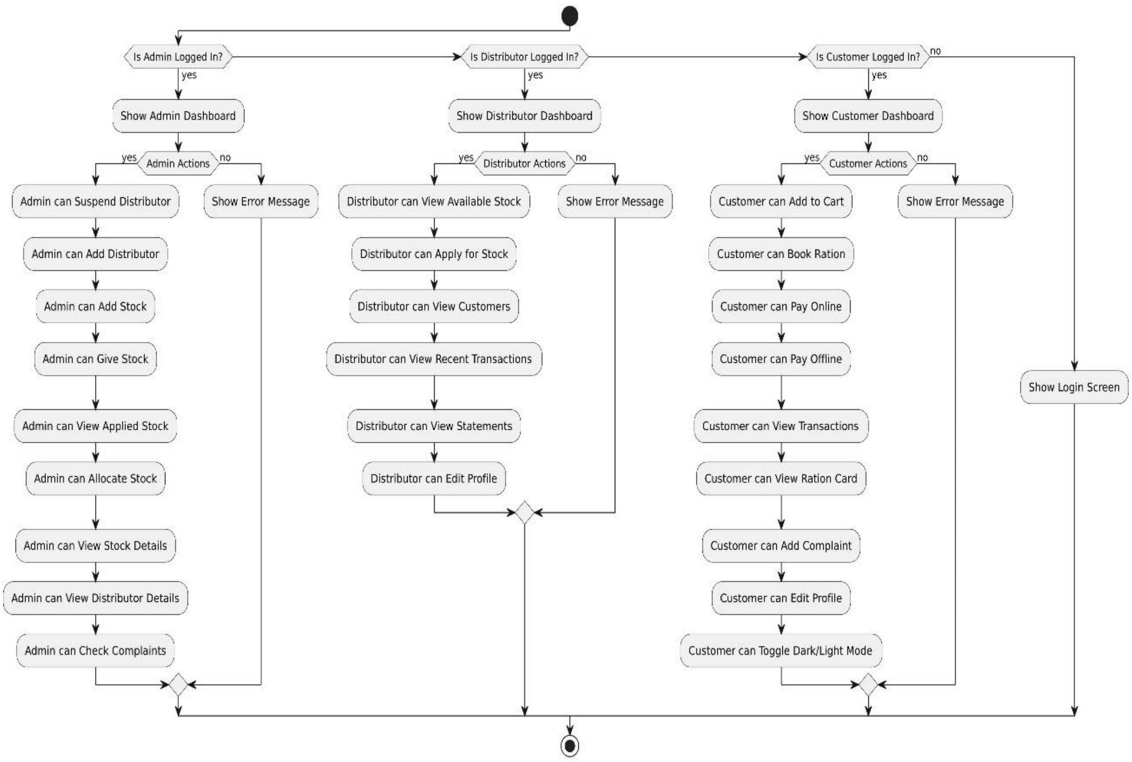

Figure 3.2.3: Activity Diagram for Digiration

3.2.4 CONTEXT DIAGRAM

The context diagram provides a high-level view of the system, showing how the Admin, Distributor, and Customer interact with the central Ration Management System. It outlines the primary data flows between users and the system, including stock management, transactions, and complaint handling.

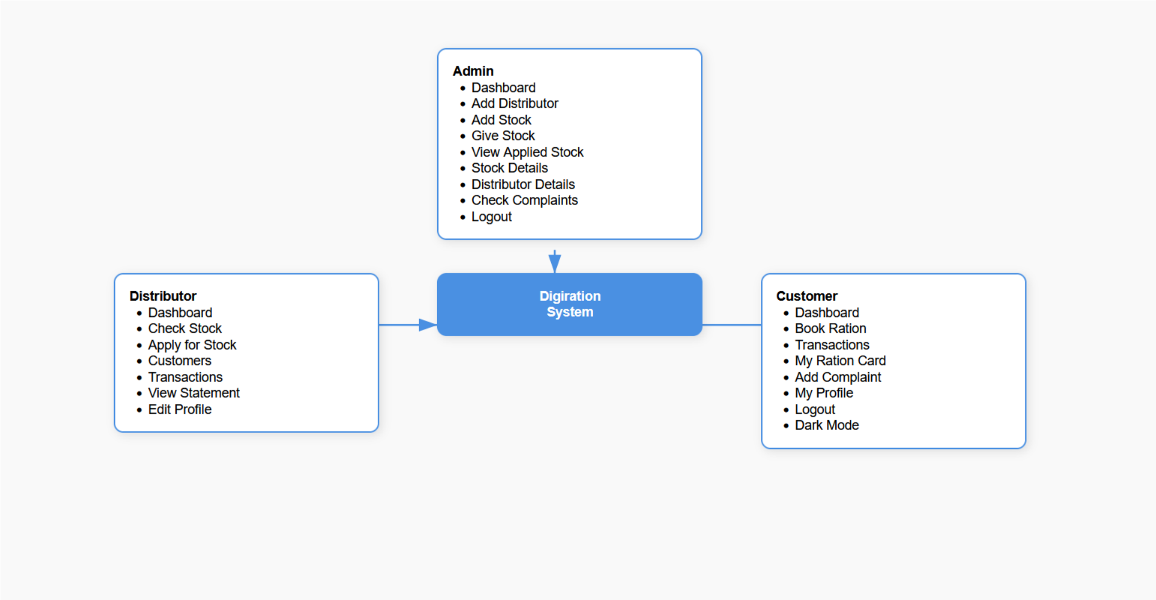

Figure 3.2.4 Context Diagram for Digiration

3.2.5 STATE DIAGRAM

The state diagram represents the different states of the system entities—Admin, Distributor, and Customer—based on their actions. It captures transitions like login, dashboard access, stock operations, booking ration, and logout, reflecting how each user progresses through various system states.

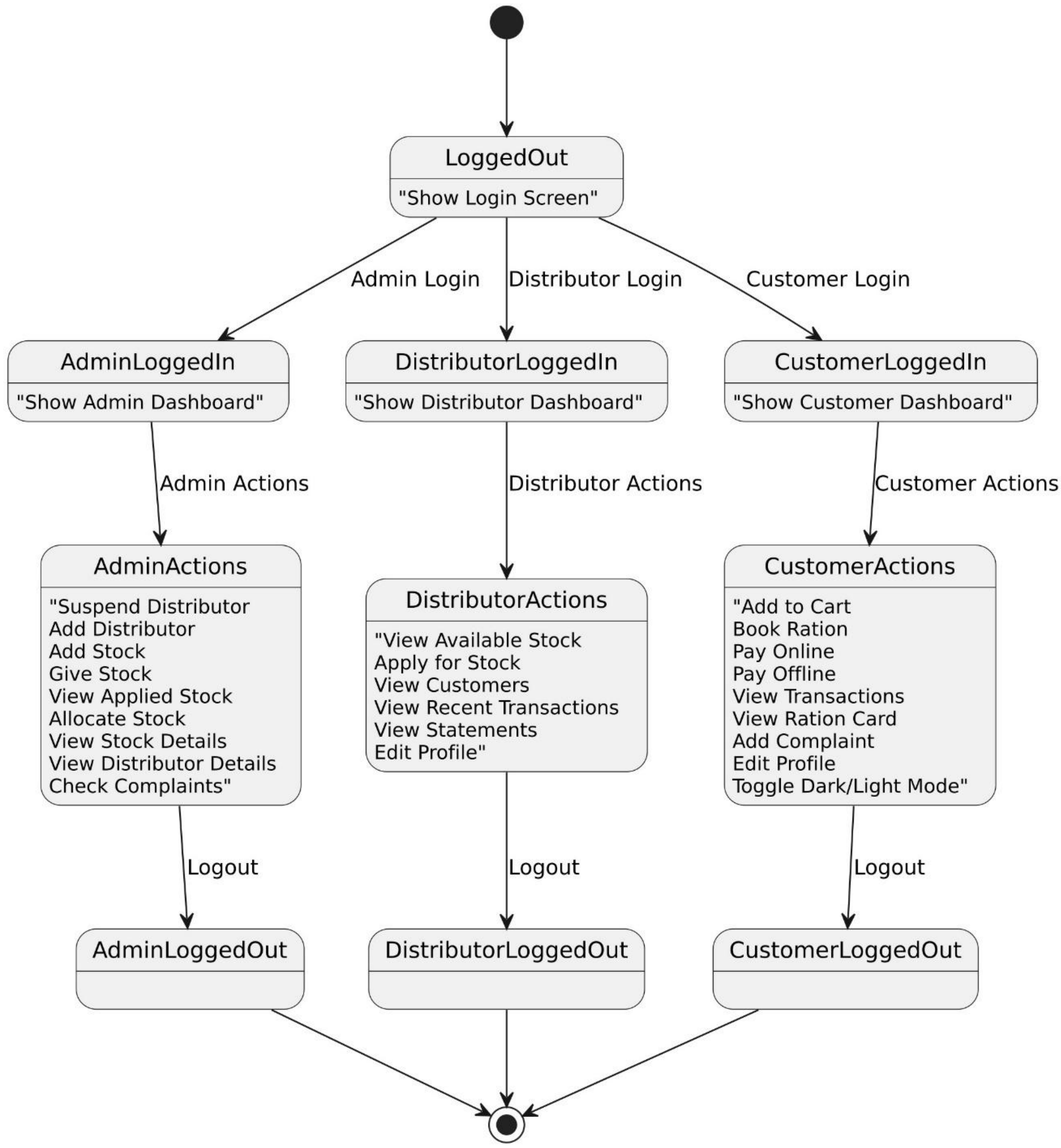

Figure 3.2.5 State diagram for Digiration

3.2.6 SEQUENCE DIAGRAM

The sequence diagram illustrates the interactions between the Admin, Distributor, and Customer in the ration distribution process. It shows the flow of actions, from stock allocation and management by the Admin to stock requests and transactions by the Distributor, and ration bookings and profile management by the Customer

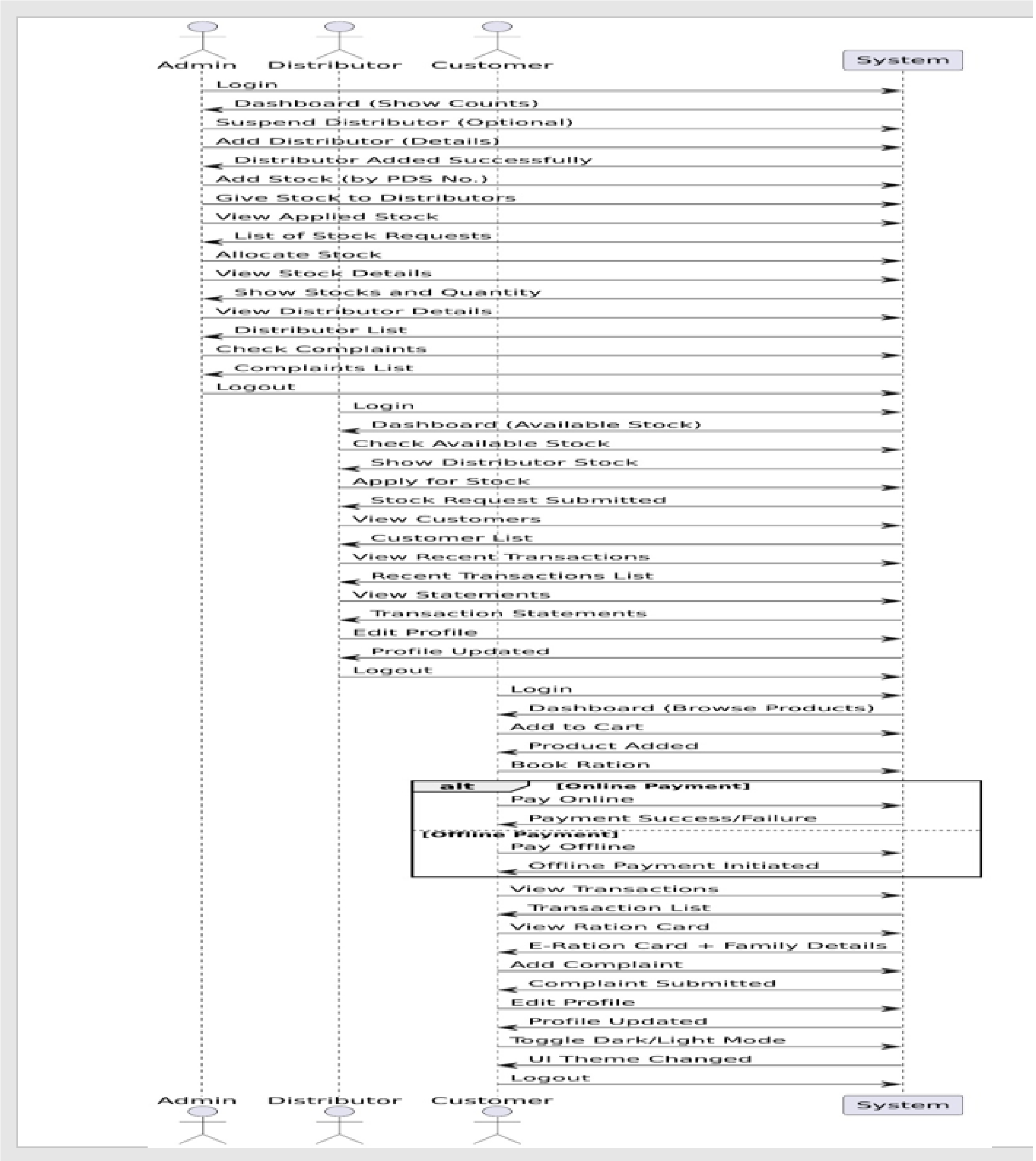

Figure 3.2.6 Sequence Diagram for Digiration

3.2.7 COMPONENT DIAGRAM

The component diagram illustrates the high-level architecture of the Ration Distribution System, showing how the Admin, Distributor, and Customer modules interact through their respective controllers and functional components. It clearly represents the system's modular structure, enabling maintainability, scalability, and role-specific operations.

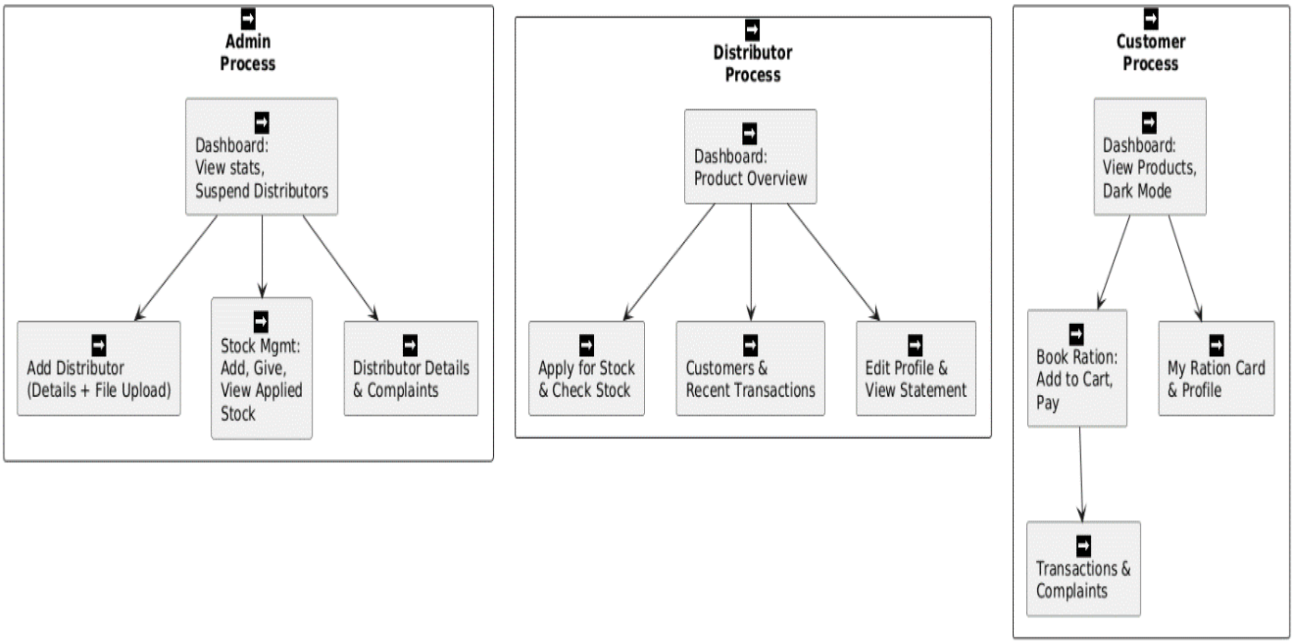

Figure 3.2.7 Component Diagram for Digiration

3.2.8 DEPLOYMENT DIAGRAM

The deployment diagram illustrates the physical architecture of the ration distribution system, showcasing how the Admin, Distributor, and Customer components are deployed across different servers. It highlights their interactions through controllers, databases, and functional modules, emphasizing system modularity and role-specific functionalities.

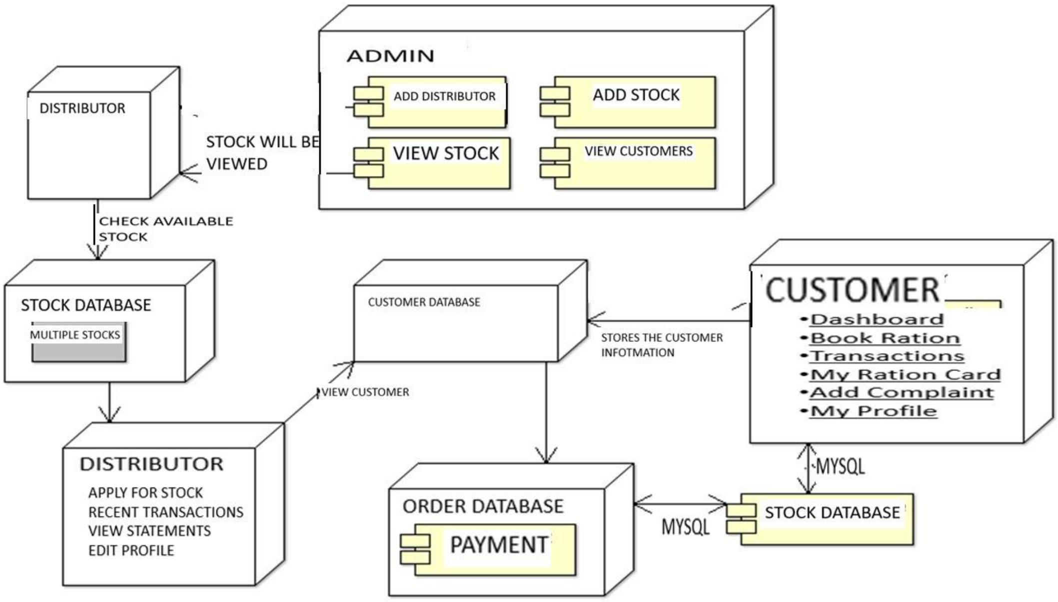

Figure 3.2.8 Deployment Diagram for Digiration

3.3 DESIGN COMPONENTS

3.2.1 Front End:

The frontend of the project is built using PHP with HTML, CSS, and JavaScript to create an interactive and user-friendly interface for Admin, Distributor, and Customer modules.

3.2.2 Back End:

The backend uses PHP for server-side logic and MySQL as the database to handle data storage, retrieval, authentication, and transaction processing.

3.4 DATABASES DESCRIPTION

Listed below gives a description of database document schemas used for Digiration

Table 3.1: Apply Stock Description

Attribute Name	Type	Width	Constraint(s)	Description
ap_id	Integer	10	Primary Key, Auto-Increment	Unique identifier for the application stock request
Date	Date	NULL	NULL	Date when the stock request is made.
stock_id	Integer	10	Foreign Key (MUL)	ID of the stock requested.
stock_price	Integer	10	NULL	Price of the requested stock.
stock_name	Varchar	255	NULL	Name of the stock requested.
d_fname	Varchar	255	NULL	First name of the distributor making the request
d_mname	Varchar	255	NULL	Middle name of the distributor making the request
d_lname	Varchar	255	NULL	Last name of the distributor making the request
Pds_no	Integer	10	NULL	Pds_no related to the request
Quantity	Integer	10	NULL	Quantity of the stock requested
Status	Varchar	255	NULL	Status of the request (approved, pending, etc.).

Table 3.2: Available Stock Description

Attribute Name	Type	Width	Constraint(s)	Description
pds_no	Integer	10	NOT NULL	Pds number related to stock
d_id	Integer	10	NOT NULL	Distributor ID related to the stock..
d_fname	Varchar	255	NULL	First name of the distributor making the request
d_mname	Varchar	255	NULL	Middle name of the distributor making the request
d_lname	Varchar	255	NULL	Last name of the distributor making the request
Stock_id	Integer	10	Foreign Key	ID of the stock available
Stock_name	Varchar	255	NOT NULL	Name of the available stock.
Quantity	Integer	10	NOT NULL	Quantity of the stock available

Table 3.3 : Book Details Description

Attribute Name	Type	Width	Constraint(s)	Description
booking_id	Integer	10	Primary Key, Auto-Increment	Unique identifier for the booking.
Rationcardno	Varchar	15	NOT NULL	Ration card number associated with the booking.
Date	Date		NULL	Date of the booking.
Amount	Varchar	5	NOT NULL	Amount to be paid for the ration
Status	Integer	11	NOT NULL	Status of the booking (0 for pending, etc.).
Items	Integer	15	NOT NULL	Number of items booked
Ration_grant	Integer	11	NOT NULL	Grant amount for the ration
Givendate	Date		NULL	Date when ration is given.

Table 3.4 Complaint Details Description

Attribute Name	Type	Width	Constraint(s)	Description
c_id	Integer	11	Primary Key, Auto-Increment	Unique identifier for the complaint.
Date	Date		NOT NULL	Date when the complaint was filed.
description	Varchar	255	NOT NULL	Detailed description of the complaint.
u_id	Integer	5	NOT NULL	User ID of the person filing the complaint.
u_fname	Varchar	255	NOT NULL	First name of the user filing the complaint.
u_mname	Varchar	255	NOT NULL	Middle name of the user.
u_lname	Varchar	255	NOT NULL	Last name of the user.
pds_no	Integer	5	NOT NULL	PDS number associated with the complaint.

Table 3.5 Distributors Details Description

Attribute Name	Type	Width	Constraint(s)	Description
d_id	Integer	11	Primary Key, Auto-Increment	Unique identifier for the distributor.
Fname	Varchar	20	NOT NULL	First name of the distributor.
Mname	Varchar	20	NOT NULL	Middle name of the distributor.
Lname	Varchar	20	NOT NULL	Last name of the distributor.
Address	Varchar	100	NOT NULL	Address of the distributor..
contact_no	Bigint	10	NOT NULL	Contact number of the distributor.
Gender	Varchar	10	NOT NULL	Gender of the distributor.
aadhar_no	Bigint	12	UNIQUE	Aadhar number of the distributor.

rationcard_no	Bigint	15	UNIQUE	Ration card number of the distributor.
pds_no	Integer	10	UNIQUE	PDS number related to the distributor.
City	Varchar	30	NOT NULL	City where the distributor is located.
State	Varchar	30	NOT NULL	State where the distributor is located.
Pincode	Integer	6	NOT NULL	Pincode of the distributor's address.
email_id	Varchar	50	NOT NULL	Email Id of the distributor
image	Varchar	254	NOT NULL	Image of the Distributor
Password	Varchar	40	NOT NULL	Password used for login by the distributor.

Table 3.6 Stock Details Description

Attribute Name	Type	Width	Constraint(s)	Description
stock_id	Integer	10	Primary Key, Auto-Increment	Unique identifier for the stock.
stock_name	Varchar	255	NOT NULL	Name of the stock item.
Quantity	Integer	10	NOT NULL	Quantity available for the stock item.
Price	Integer	10	NOT NULL	Price per unit of the stock item.
description	Varchar	255	NULL	Description of the stock item.

Table 3.7 User Details Description

Attribute Name	Type	Width	Constraint(s)	Description
u_id	Integer	10	Primary Key, Auto-Increment	Unique identifier for the user.
Fname	Varchar	50	NOT NULL	First name of the user.
Lname	Varchar	50	NOT NULL	Last name of the user.

Email	Varchar	255	NOT NULL	Email address of the user.
Password	Varchar	255	NOT NULL	Password for user login.
contact_no	Varchar	15	NULL	Contact number of the user.
Address	Varchar	255	NULL	Address of the user.
Role	Varchar	20	NOT NULL	Role of the user (Admin,Distributor,etc.)

Table 3.8 Payment Details Description

Attribute Name	Type	Width	Constraint(s)	Description
payment_id	Integer	10	Primary Key, Auto-Increment	Unique identifier for the payment transaction.
booking_id	Integer	10	Foreign Key (MUL)	Reference to the related booking.
Amount	Integer	10	NOT NULL	Amount to be paid in the transaction.
Payment_status	Varchar	20	NOT NULL	Status of the payment
payment_date	Date		NOT NULL	Date when the payment was made.
payment_method	Varchar	20	NOT NULL	Method of payment used (Cash, Credit, etc.).

Table 3.9 Distributor Recipt Details

Attribute Name	Type	Width	Constraint(s)	Description
receipt_id	Integer	11	Primary Key, Auto-Increment	Unique identifier for each receipt.
booking_id	Integer	10	Foreign Key (MUL)	Refers to the booking associated with this receipt.
Date	Date		NOT NULL	Date of the receipt generated
Rationcard_no	Varchar	20	NOT NULL	Ration number of the customer
d_pds	Varchar	100	NOT NULL	Details of items under Public Distribution System
Amount	Decimal	10,2	NOT NULL	Total amount for the distributed items.

Table 3.10 Give Stock Details

Attribute Name	Type	Width	Constraint(s)	Description
Id	Integer	11	Primary Key, Auto-Increment	Unique identifier for each stock entry.
pds_no	Varchar	20	NOT NULL	PDS number to which the stock is given.
stock_name	Varchar	100	NOT NULL	Name of the stock item.
Quantity	Integer	10	NOT NULL	Quantity of the stock given

Table 3.11: Pds Stock details

Attribute Name	Type	Width	Constraint(s)	Description
pds_no	Varchar	20	Primary Key,NOT NULL	Unique identifier for each PDS entry.
stock_name	Varchar	100	NOT NULL	Name of the stock item.
Quantity	Integer	11	NOT NULL	Quantity of the stock Available

Table 3.12 Ration Member Details

Attribute Name	Type	Width	Constraint(s)	Description
rationcard_no	Varchar	20	Primary Key, Auto-Increment	Unique number assigned to the ration card.
m_name	Varchar	20	NOT NULL	Name of the main member(head of the family).
m_age	Integer	10	NOT NULL	Age of the main member.
m_aadhar	Varchar	12	NOT NULL	Aadhar number of the main member
P1_name	Varchar	100	NOT NULL	Name of the family member
P1_age	Bigint	10	NOT NULL	Age of the family member
P1_aadhar	Varchar	12	NOT NULL	Aadhar number of the family member
f_members	Integer	2	NOT NULL	Total number of family members.
phone_number	Varchar	10	Unique,NOT NULL	Registered phone number of the family/ration card holder.

Table 3.13 Receipt Details

Attribute Name	Type	Width	Constraint(s)	Description
receipt_id	Varchar	20	Primary Key, Auto-Increment	Unique ID for each receipt. .
u_name	Varchar	20	NOT NULL	Name of the User
u_contact_no	Varchar	10	NOT NULL	User's Contact Number
Amount	Decimal	10,2	NOT NULL	Amount paid in the transaction
Rationcard_no	Varchar	20	Foreign key	Linked RationCard Number
Date	Date		NOT NULL	Date of receipt generation.
Time	Time		NOT NULL	Time of the transaction
Tid	Varchar	20		Transaction ID or status
state	Integer	`1		Status of the transaction (e.g., 0 for pending).
d_id	Integer	10	Foreign Key	Distributor ID who processed the transaction.
booking_id	Integer	10	Foreign Key	Booking ID related to the receipt.

Table 3.14 Send Request Details

Attribute Name	Type	Width	Constraint(s)	Description
Id	Integer	10	Primary Key, Auto-Increment	Unique ID for each request.
Name	Varchar	20	NOT NULL	Name of the Requester
Date	Date		NOT NULL	Date the Request was sent
Email	Varchar	100	NOT NULL	Email address of the requester
Subject	Varchar	100	NOT NULL	Subject or title of the request.
Message	Text		NOT NULL	Full message content of the request.

3.5 USER INTERFACE DESIGN

Figure 3.5.1 provide the interface for main activity.

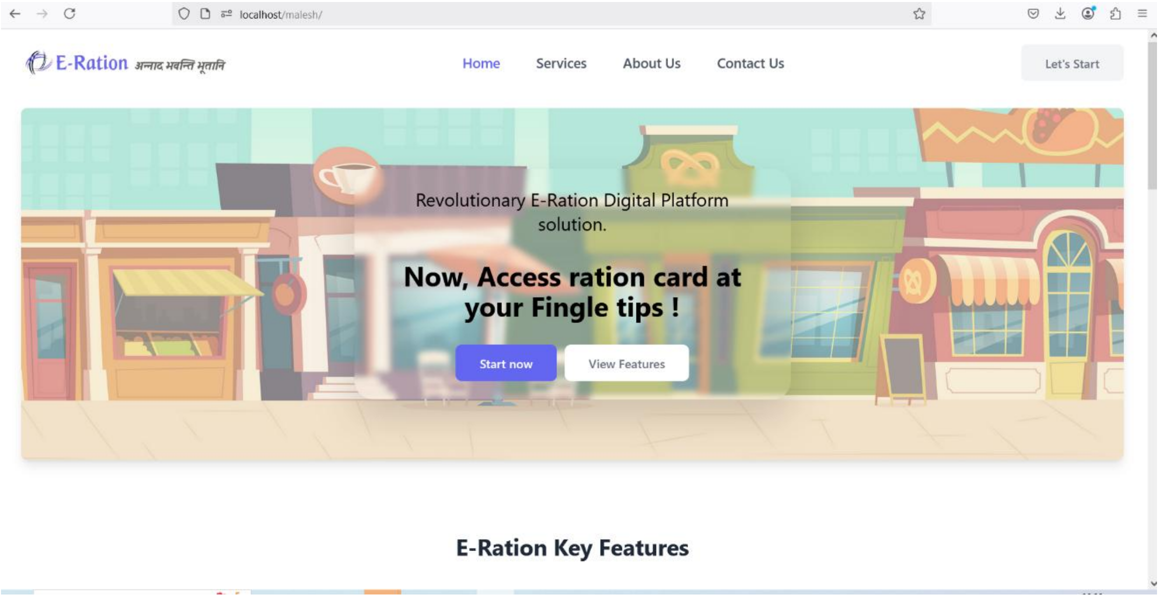

Figure 3.5.1: Main layout of Digiration

Figure 3.5.2 provide the interface for Login page

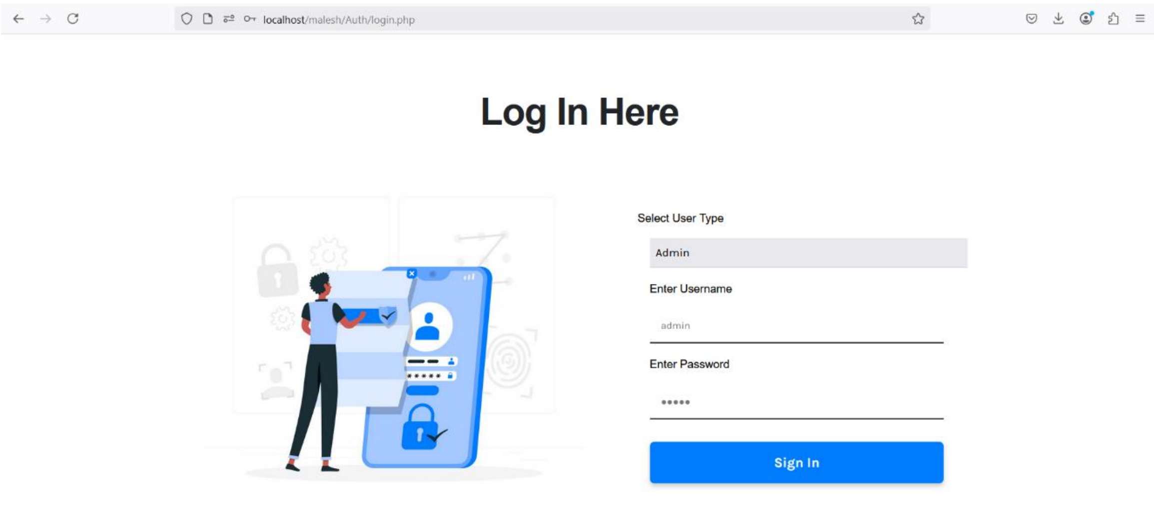

Figure 3.5.2: Login Page

Now all the figures provide the interface for Admin

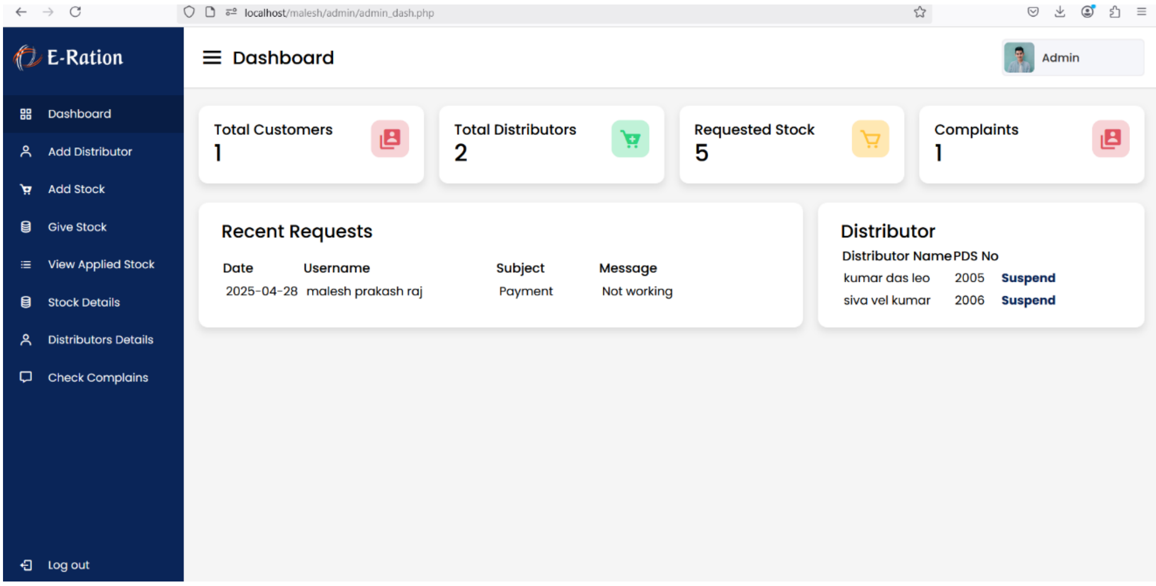

Figure 3.5.3 Dashboard of the Admin

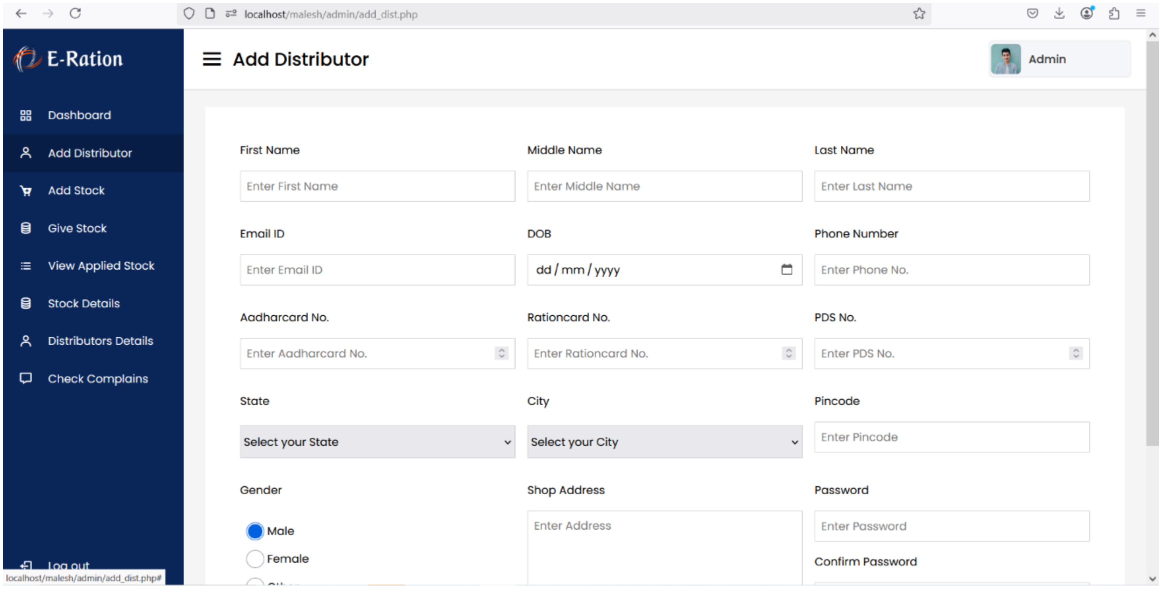

Figure 3.5.4: Add Distributer page for Admin

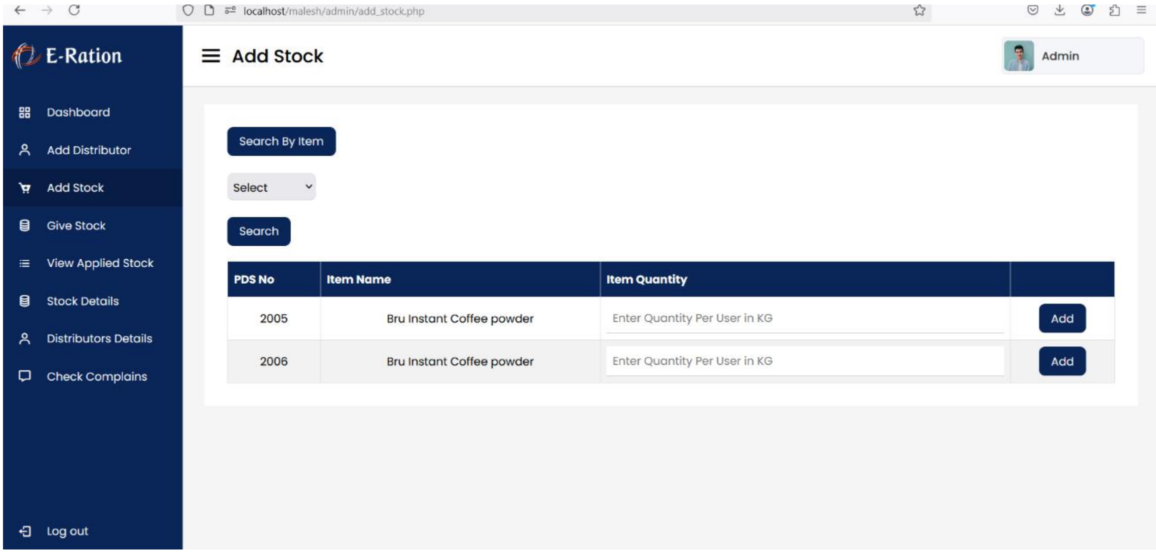

Figure 3.5.5: Add Stock page for Admin

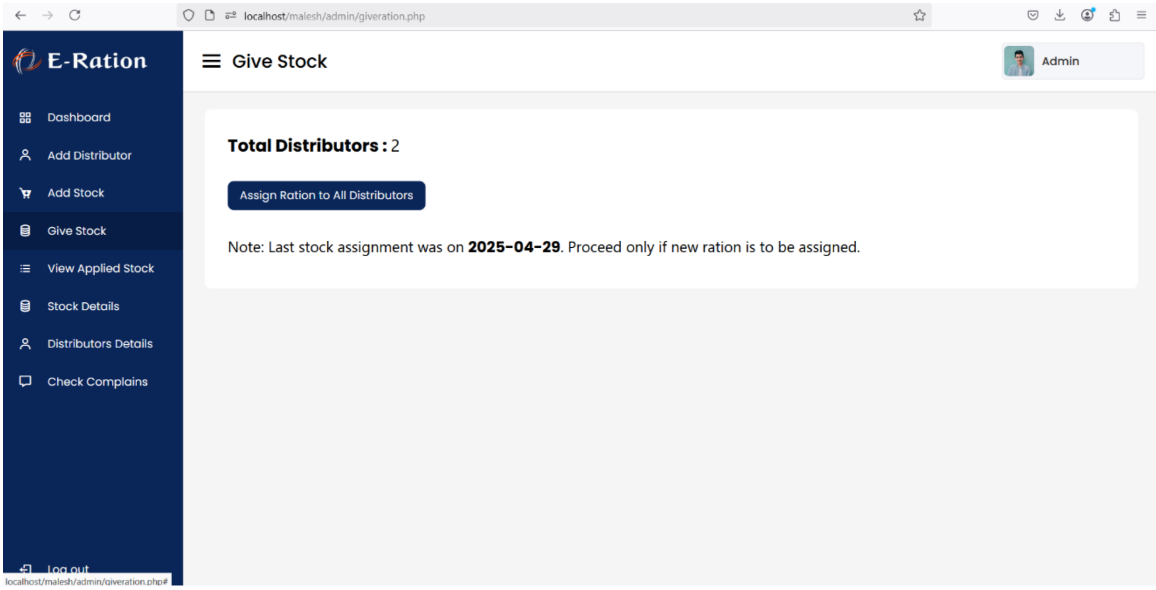

Figure 3.5.6: Give stock page for Admin

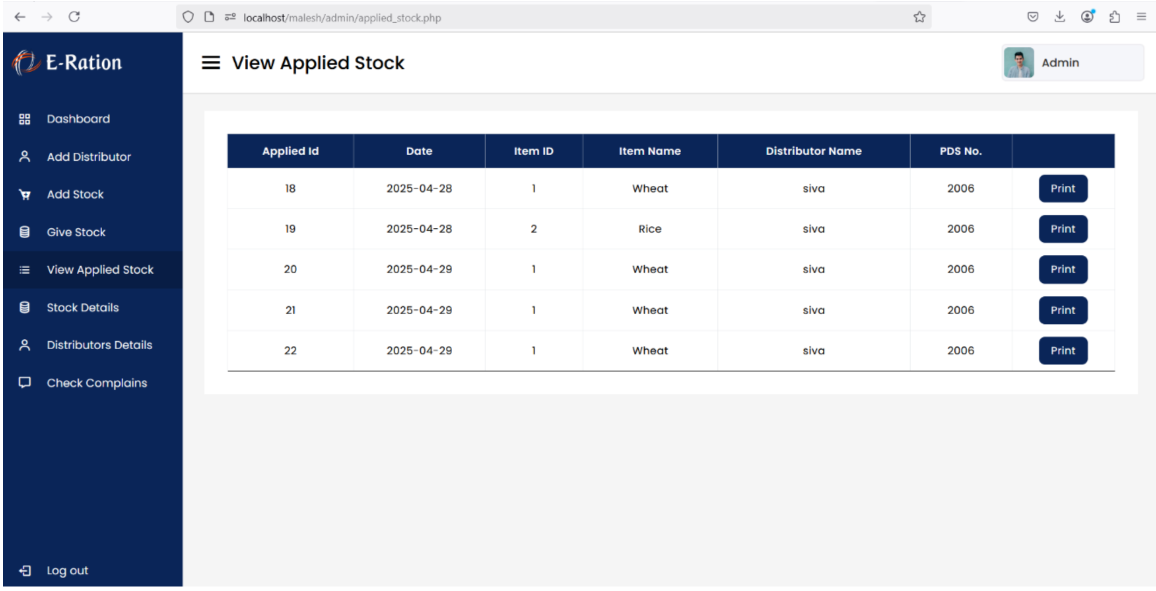

Figure 3.5.7: View Applied Stock for Admin

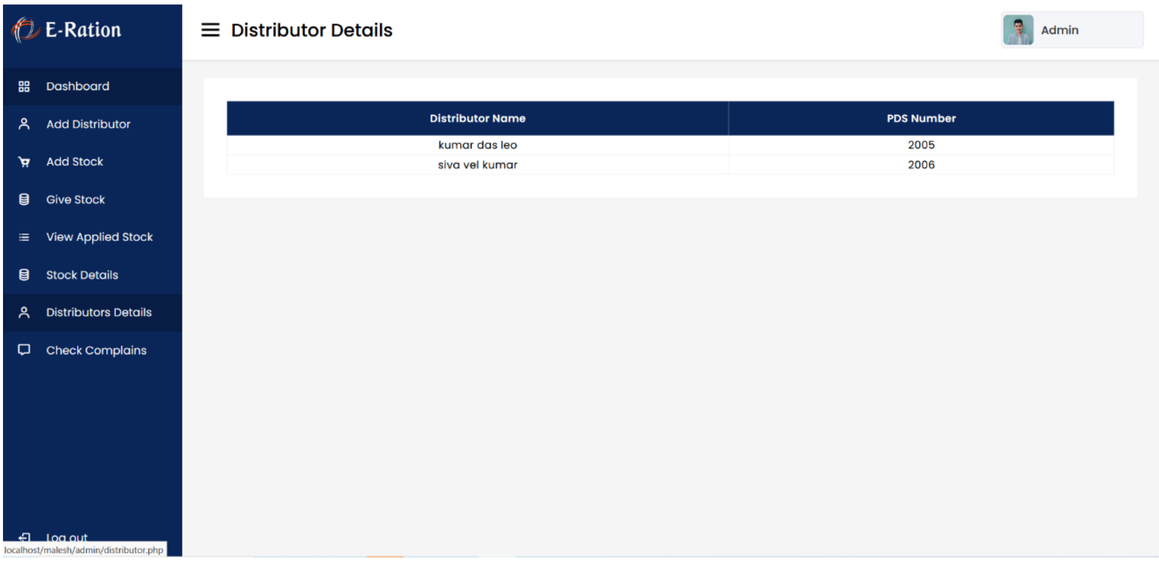

Figure 3.5.8: Distributor Details for Admin

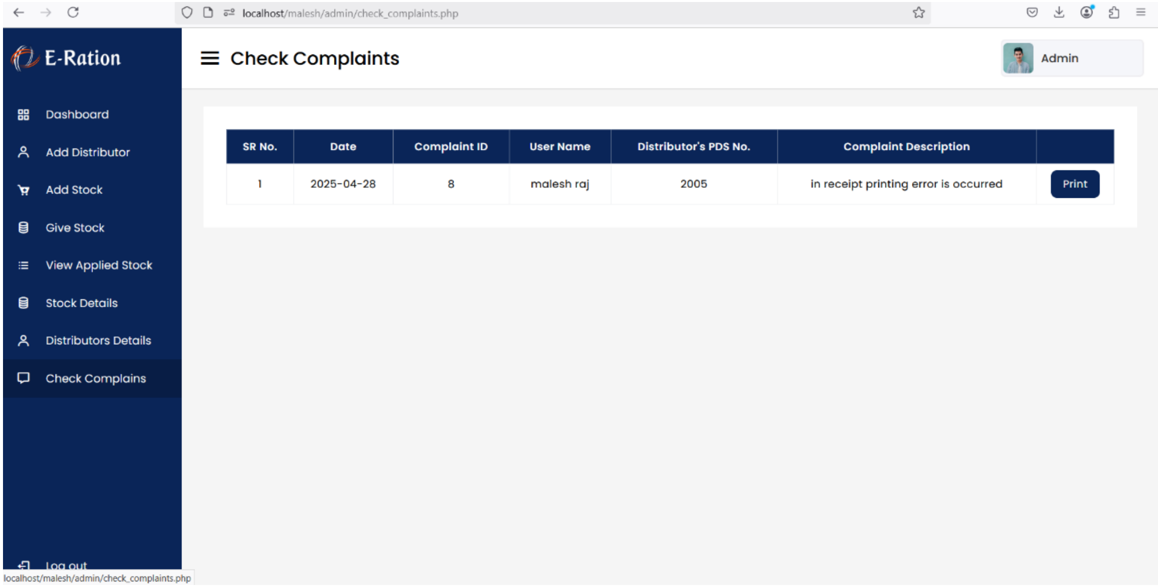

Figure 3.5.9: Check complaints page for Admin

Now all the figures provide the interface for Distributor

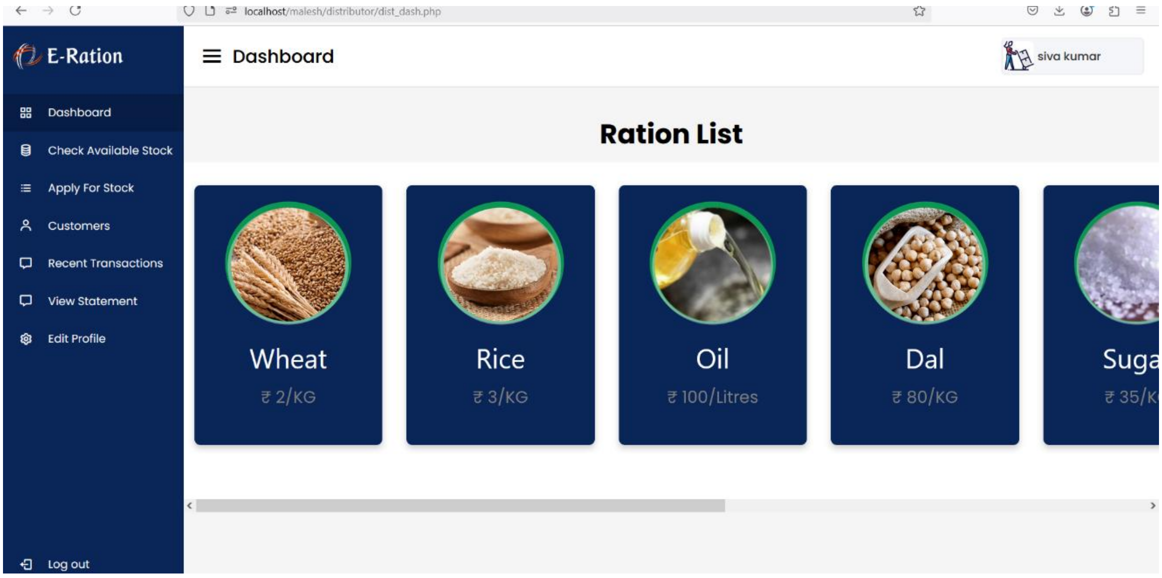

Figure 3.5.10: Dashboard for Distributor

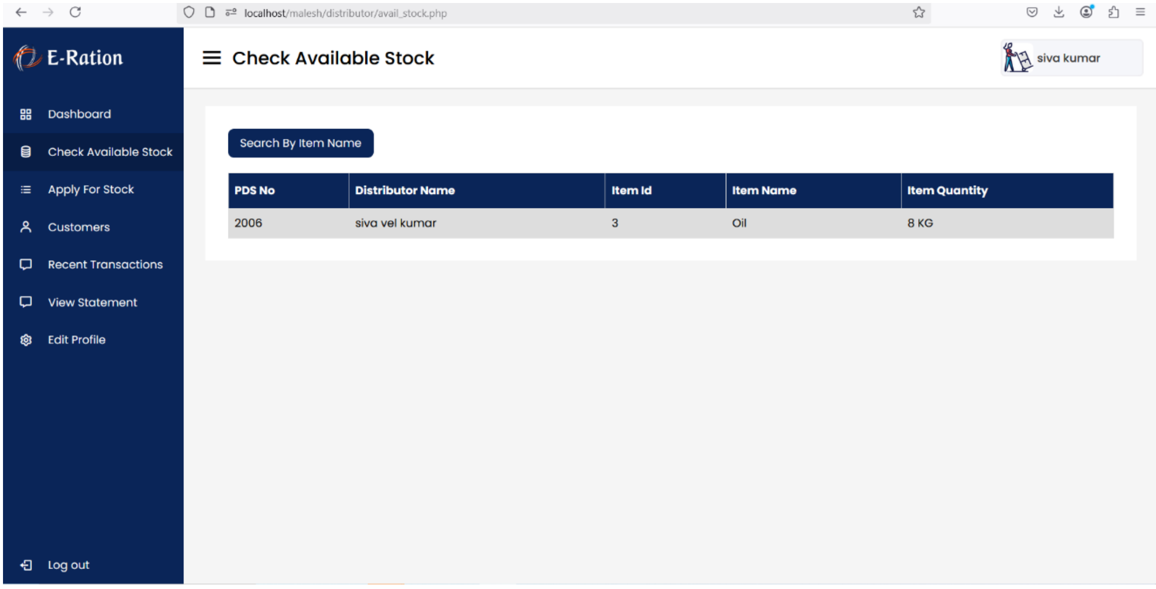

Figure 3.5.11: Check Available page for Distributor

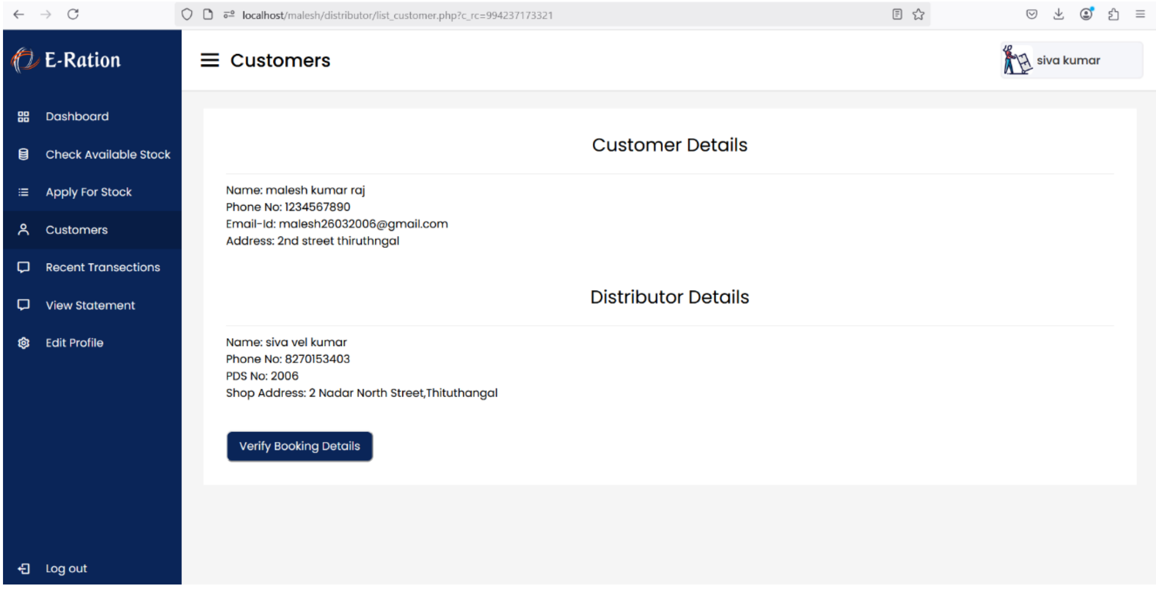

Figure 3.5.12: Customers page for Distributor

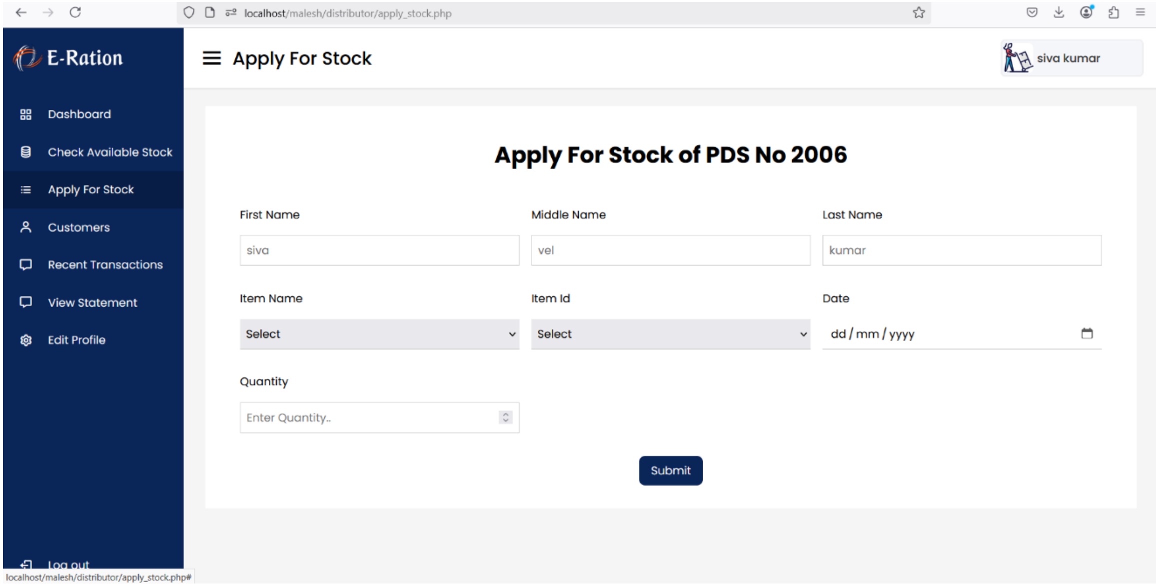

Figure 3.5.13 Apply Stock Page for Distributor

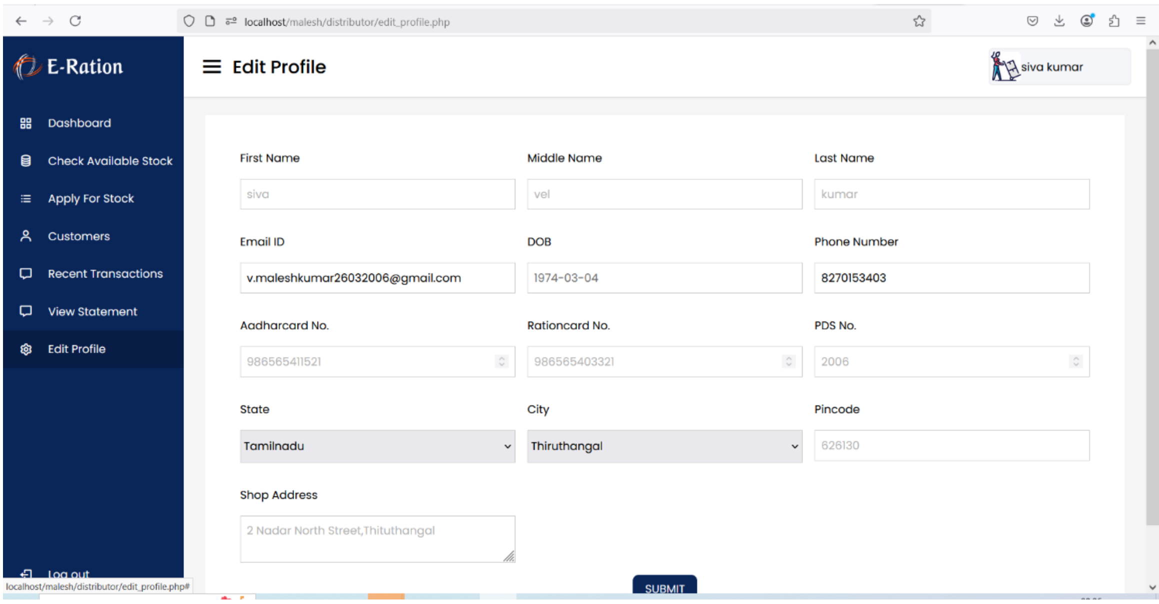

Figure 3.5.14 Edit Profile Page for Distributor

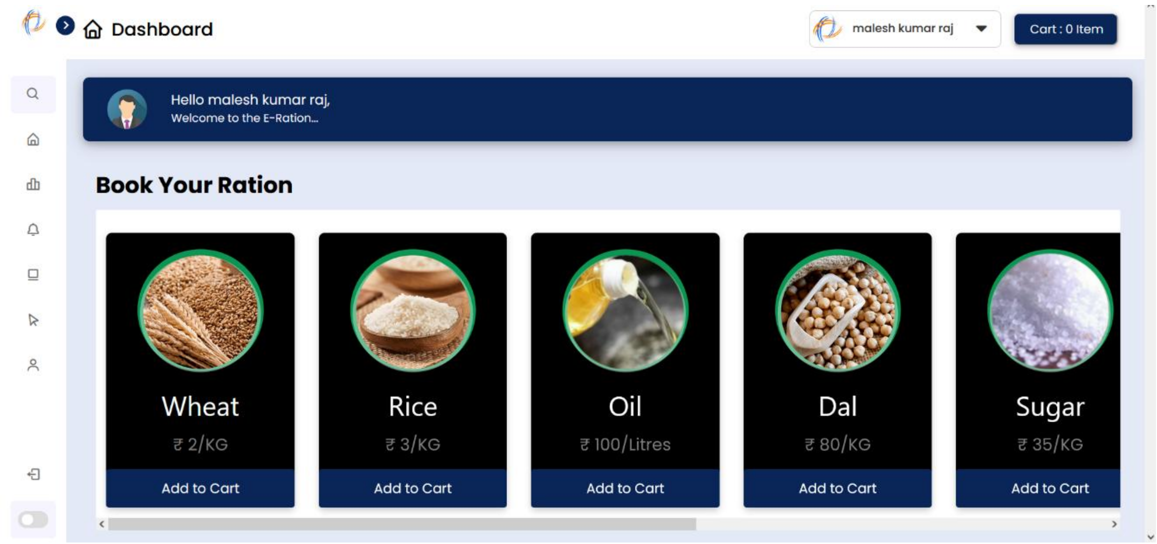

Figure 3.5.15 Dashboard Page for Customer

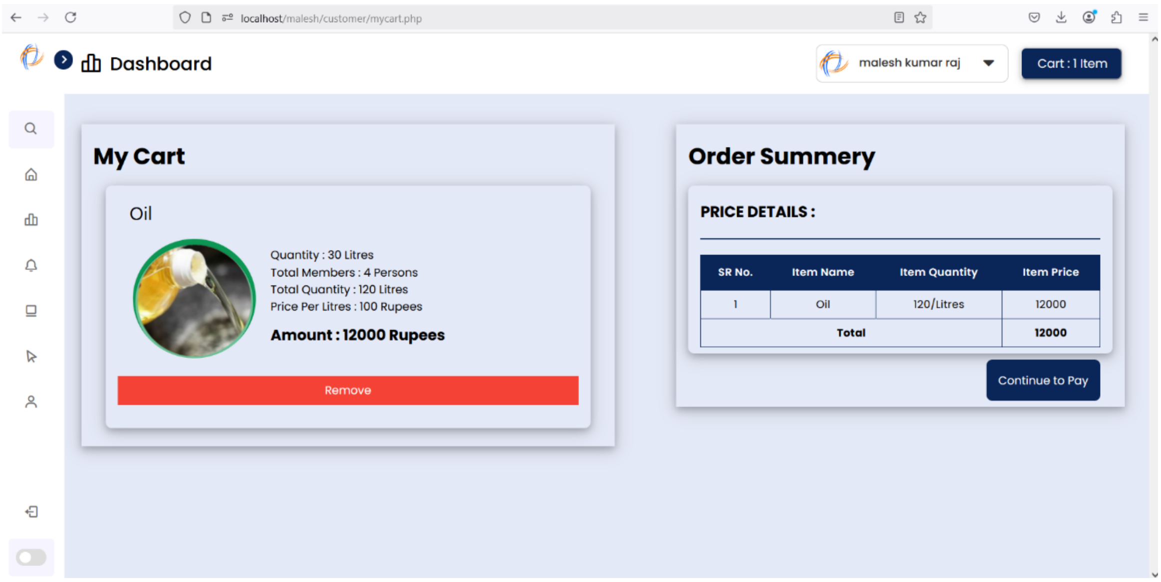

Figure 3.5.16 My Cart page for Customer

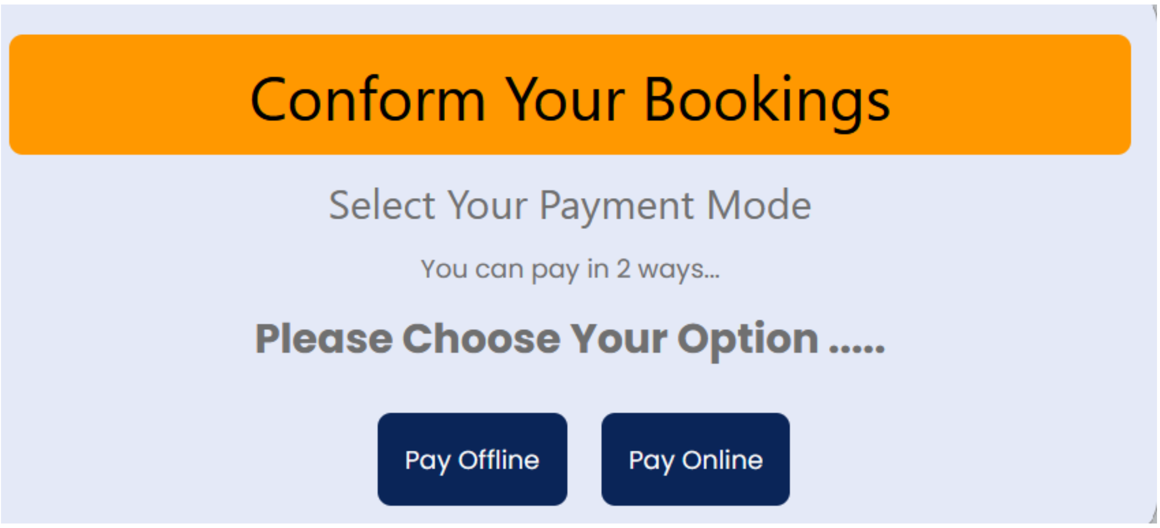

Figure 3.5.17 Payment Page for Customer

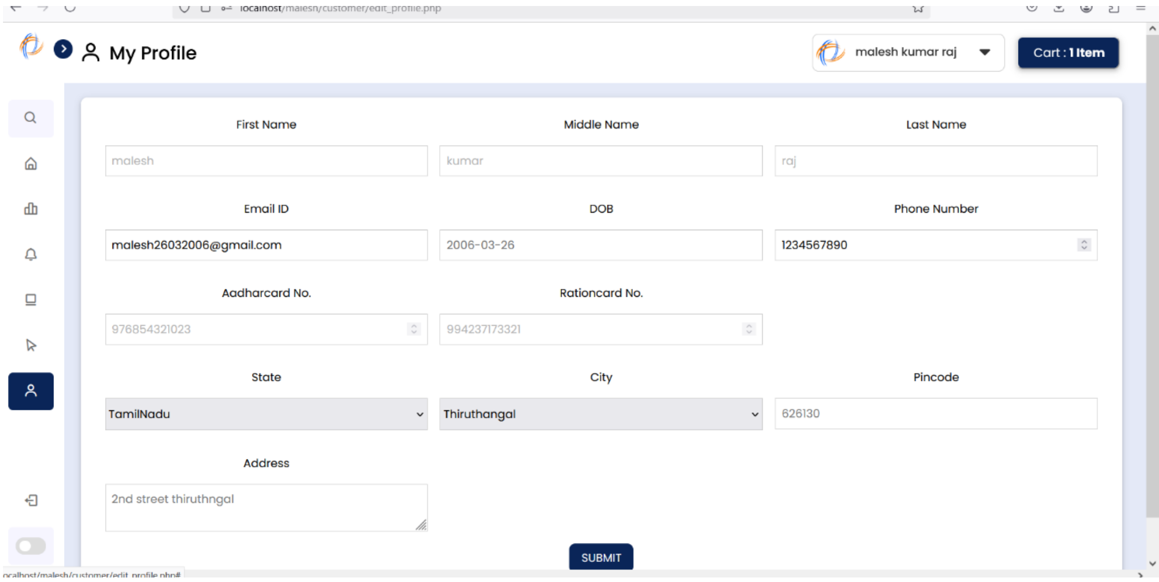

Figure 3.5.18 Profile Page for Customer

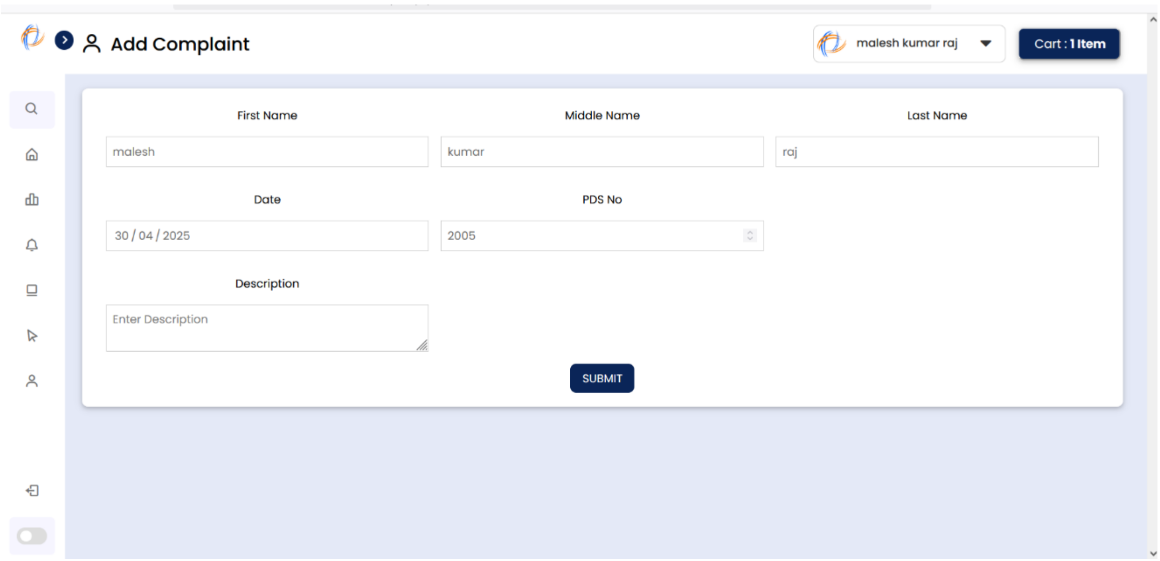

Figure 3.5.19 Add Complaint page for Customer

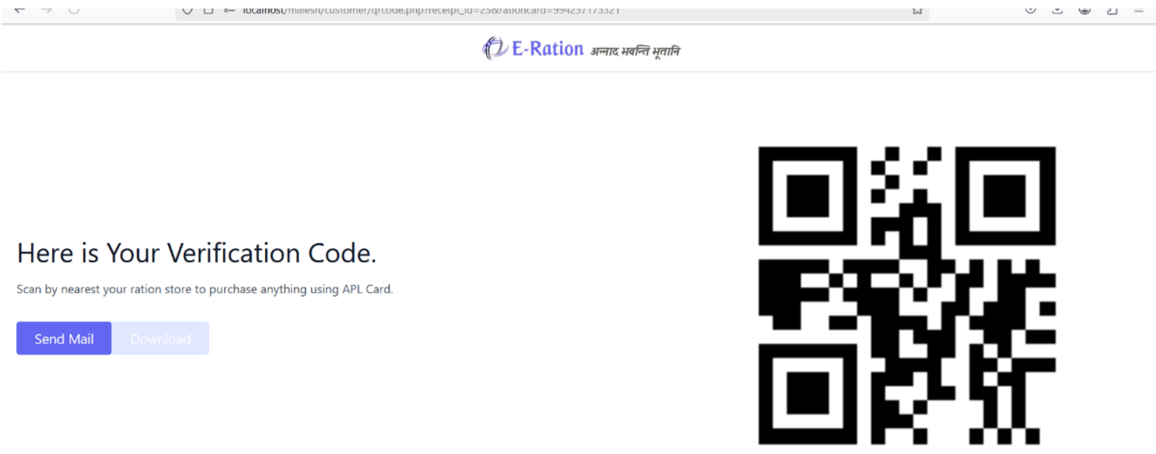

Figure 3.5.20 Verification QR Code for Offline pay

Conclusion

The proposed system offers an efficient, transparent, and user-friendly digital solution for managing ration distribution. It streamlines operations for admins, distributors, and customers while ensuring accountability and accessibility. This model significantly improves upon existing manual processes, fostering a more reliable public distribution ecosystem.

CHAPTER 4

SYSTEM IMPLEMENTATION

4.1 LOGIN IMPLEMENTATION

The login credentials are verified from the database. If correct, the user is redirected to the respective dashboard based on role (Admin, Distributor, Customer).

GET username, password
IF username, password valid
→ RETURN corresponding user dashboard
ELSE
→ TOAST Invalid Credentials

4.2 ADMIN ADD DISTRIBUTOR IMPLEMENTATION

Admin fills the distributor registration form. The system validates each field and uploads the file. If successful, the distributor is added to the system.

GET all distributor form fields
IF all fields valid AND file uploaded
→ RETURN "Distributor added successfully"
ELSE
→ TOAST Enter valid details

4.3 ADD STOCK IMPLEMENTATION (Admin)

Admin selects the distributor using the PDS number and adds stock quantities. Stock is updated in the distributor's profile.

GET PDS number, stock details
IF PDS number exists
→ UPDATE distributor's stock
ELSE
→ TOAST Invalid PDS Number

4.4 GIVE STOCK TO ALL DISTRIBUTORS (Admin)

Admin clicks “Assign Ration to All,” and the stock is distributed equally or proportionally to active distributors.

GET all active distributors

CALCULATE stock distribution

→ UPDATE stock for each distributor

RETURN success message

4.5 VIEW APPLIED STOCK (Admin)

System fetches stock requests from all distributors. Admin reviews and decides to allocate.

GET stock applications

IF requests exist

→ DISPLAY applied requests

ELSE

→ TOAST No current stock requests

4.6 STOCK DETAILS (Admin)

Admin views total stock quantities available across different categories.

GET stock inventory

→ DISPLAY in dashboard/table view

4.7 DISTRIBUTOR DETAILS (Admin)

Admin fetches all registered distributor profiles and their activity status.

GET distributor database

→ DISPLAY name, contact, status

OPTION Suspend if inactive

4.8 CHECK COMPLAINTS (Admin)

All customer complaints are fetched and shown to admin with filters like date, distributor, type.

GET complaints

→ DISPLAY to admin

OPTION Mark as resolved

4.9 DISTRIBUTOR DASHBOARD IMPLEMENTATION

Distributor dashboard fetches available stock, customer count, and notifications from admin.

GET stock, customers, messages

→ DISPLAY in widgets and tables

4.10 CHECK AVAILABLE STOCK (Distributor)

Distributor checks current available inventory.

GET from inventory

→ DISPLAY with item name and quantity

4.11 APPLY FOR STOCK (Distributor)

Distributor selects items and quantity, then submits a stock request to admin.

GET selected items, quantity

→ SEND stock application to admin

→ TOAST Request submitted successfully

4.12 CUSTOMERS VIEW (Distributor)

Distributor views customers registered under their shop/distributor ID.

GET distributor ID

→ DISPLAY customer list

4.13 RECENT TRANSACTIONS (Distributor)

Shows transaction history between the distributor and the system.

GET transaction logs

→ DISPLAY by date

4.14 VIEW STATEMENT (Distributor)

Distributor selects date range to view financial or supply statement.

GET start date, end date

→ DISPLAY filtered transaction statement

4.15 EDIT PROFILE (Distributor)

Distributor edits their details and uploads necessary files.

GET updated fields

IF fields valid

→ UPDATE database

→ TOAST Profile updated

4.16 CUSTOMER DASHBOARD IMPLEMENTATION

Displays available products, ration balance, and distributor info.

GET product list, e-ration, distributor info

→ DISPLAY in dashboard

4.17 BOOK RATION (Customer)

User adds items to cart and proceeds with offline/online payment.

GET cart items, payment method

IF payment successful

→ UPDATE transaction

→ TOAST Booking confirmed

ELSE

→ TOAST Payment failed

4.18 VIEW TRANSACTIONS (Customer)

User sees history of completed and failed transactions.

GET user ID

→ DISPLAY transactions with status

4.19 MY RATION CARD (Customer)

Displays e-ration card with personal, family, and distributor details.

GET user ID

→ DISPLAY ration card data

4.20 ADD COMPLAINT (Customer)

User enters complaint subject and message, submits to admin.

GET complaint form

IF valid input

→ STORE in complaints DB

→ TOAST Complaint submitted

4.21 EDIT PROFILE (Customer)

Customer updates profile information.

GET updated form

IF valid

→ UPDATE database

→ TOAST Profile updated

4.22 DARK MODE TOGGLE (Customer)

User toggles between dark and light mode.

GET theme state

→ UPDATE UI based on toggle

CHAPTER 5

RESULTS AND DISCUSSION

5.1 TEST CASES AND RESULTS

This shows that the possible test data for the both positive and negative test case given below, if the user is already having account then the output is true otherwise false.

5.1.1 Test Cases and Results for Login Function

Test Case ID	Test Case Description	Test Data	Expected Output	Result
TC1	Valid login details	Username:admin Password:admin	TRUE	PASS
TC2	Invalid login with incorrect phone	Username:admin1 Password:admin	FALSE	PASS

5.1.2 Test Cases and Results for Admin – Add Distributor

Test Case ID	Test Case Description	Test Data	Expected Output	Result
TC3	All valid fields with file upload	Valid form data, valid file	Distributor added	PASS
TC4	Missing required fields	Missing Phone Number	Validation Error	PASS
TC5	Invalid Aadhar number format	Aadhar: 1234	Validation Error	PASS

5.1.3 Test Cases and Results for Admin – Add Stock

Test Case ID	Test Case Description	Test Data	Expected Output	Result
TC6	Valid distributor PDS number and quantity	PDS123456, Rice: 50kg	Stock added	PASS
TC7	Invalid PDS number	PDS000000	Error Message	PASS

5.1.4 Test Cases for Admin – Give Stock to All Distributors

Test Case ID	Test Case Description	Test Data	Expected Output	Result
TC8	Trigger mass distribution with stock available	Button Click	Stock distributed	PASS
TC9	No distributors registered	Empty List	No distribution	PASS

5.1.5 Test Cases for Admin – View Applied Stock

Test Case ID	Test Case Description	Test Data	Expected Output	Result
TC10	Distributor has applied for stock	D12345	Display request	PASS
TC11	No requests made	—	No stock applications	PASS

5.1.6 Test Cases for Admin – View Complaints

Test Case ID	Test Case Description	Test Data	Expected Output	Result
TC12	Fetching customer complaints	C123	Display complaints	PASS
TC13	No complaints in system	—	"No complaints" message	PASS

5.1.7 Test Cases for Distributor – Apply for Stock

Test Case ID	Test Case Description	Test Data	Expected Output	Result
TC14	Valid request for stock	Rice: 20kg	Request Submitted	PASS
TC15	Apply with empty quantity	Rice: 0kg	Validation error	PASS

5.1.8 Test Cases for Distributor – Edit Profile

Test Case ID	Test Case Description	Test Data	Expected Output	Result
TC16	Edit valid name, email	Name: Raju, Email: x@x.com	Profile updated	PASS
TC17	Edit with invalid email	Email: wrongformat	Validation error	PASS

5.1.9 Test Cases for Customer – Book Ration

Test Case ID	Test Case Description	Test Data	Expected Output	Result
TC18	Add to cart and pay online	Rice, Oil – ₹500, UPI selected	Booking Confirmed	PASS
TC19	Payment failed during booking	Timeout or failed transaction	Payment failed	PASS

5.1.10 Test Cases for Customer – Add Complaint

Test Case ID	Test Case Description	Test Data	Expected Output	Result
TC20	Valid subject and message	"Delay", "Not received ration"	Complaint Submitted	PASS
TC21	Blank subject or message	""	Validation Error	PASS

5.1.11 Test Cases for Customer – Dark Mode

Test Case ID	Test Case Description	Test Data	Expected Output	Result
TC22	Toggle to dark mode	Switch: ON	UI switches to dark	PASS
TC23	Toggle back to light mode	Switch: OFF	UI switches to light	PASS

CHAPTER 6

CONCLUSION AND FUTURE ENHANCEMENT

The Rationing System is a comprehensive platform designed to streamline ration stock distribution and management, catering to three primary user roles: Admin, Distributor, and Customer. The Admin has full control over the system, managing tasks like monitoring customers, complaints, distributors, and stock requests, while also suspending non-compliant distributors. They can add distributors, allocate stock, and handle customer complaints. Distributors are able to manage stock, request more when necessary, track customer details, view recent transactions, and access their transaction statements, ensuring smooth operations at the distribution level. Customers can view available stock, add items to their cart, book rations, and pay either offline or online. They can also track their transaction history, maintain an e-ration card, file complaints, and switch between dark and light modes for accessibility. This system enhances transparency, efficiency, and accountability in ration distribution.

For future enhancements, the system can expand by developing a mobile app for easier access, integrate advanced analytics for better decision-making, and implement AI-based stock management to predict stock shortages. A complaint escalation system can ensure quicker resolutions, while multi-language support would cater to diverse users. Government integration can help streamline the verification process, and enhanced payment gateway options can provide more flexibility for customers. An AI chatbot can improve customer service, and IoT integration could ensure real-time stock monitoring. A smart notification system would keep users informed about updates, further enhancing user interaction. These features would elevate the platform, making it more efficient, user-friendly, and automated.

APPENDIX – A

SYSTEM REQUIREMENTS

HARDWARE REQUIREMENTS:

Processor :	Any modern processor (e.g., Intel i3 or above)
RAM :	Minimum 4 GB (8 GB recommended)
HDD :	Minimum 500 GB HDD or 256 GB SSD

SOFTWARE REQUIREMENTS:

Operating System :	Any (Windows, Linux, macOS)
DBMS :	MySQL
Web Server :	Apache (via XAMPP, WAMP, or LAMP stack)
Scripting Language:	PHP

APPENDIX – B

SOURCE CODE

```
<?php
session_start();
if (isset($_SESSION['user'])) {
?><?php
    include '../php/config.php';
    //This script will handle login
    //session_start();
  /* check if the user is already logged in
    if(isset($_SESSION['username']))
    {
      header("location: distributor.php");
      exit;
    }*/
    include "../php/connection.php";


    if (isset($_POST['btn-add-dist'])) {
        include "../php/connection.php";
        $filename = $_FILES["uploadfile"]["name"];
        $tempname = $_FILES["uploadfile"]["tmp_name"];
        $folder = "../uploads_images/";
        $post_password1 = md5(mysqli_real_escape_string($conn, $_POST['password']));
        $post_password2 = md5(mysqli_real_escape_string($conn,
$_POST['con_password']));
        if ($post_password1 == $post_password2) {
            $post_fname = $_POST['first_name'];
            $post_mname = $_POST['middle_name'];
```

```
$post_lname = $_POST['last_name'];
$post_address = $_POST['address'];
$post_ph_no = $_POST['phone_no'];
$post_gen = $_POST['gender'];
$post_dob = $_POST['dob'];
$post_email = $_POST['email'];
$post_ac_no = $_POST['acard_no'];
$post_rc_no = $_POST['rcard_no'];
$post_city = $_POST['city'];
$post_state = $_POST['state'];
$post_pincode = $_POST['pincode'];
$post_pds_no = $_POST['pds_no'];
$post_email = $_POST['email'];
$sql = "INSERT INTO tbl_distributor
(fname,mname,lname,address,contact_no,gender,dob,aadhar_no,rationcard_no,pds_no,cit
y,state,pincode,email_id,image,password)
VALUES ('$post_fname', '$post_mname', '$post_lname', '$post_address',
'$post_ph_no', '$post_gen', '$post_dob', '$post_ac_no', '$post_rc_no',
'$post_pds_no', '$post_city', '$post_state', '$post_pincode', '$post_email',
'$filename','$post_password1')";
$sql2 = "SELECT * FROM tbl_stock";
$result2 = mysqli_query($conn, $sql2);
$stocks = mysqli_fetch_all($result2, MYSQLI_ASSOC);
foreach ($stocks as $stock) {
    $stock_name = $stock['stock_name'];
    if ($stock_name == "Wheat") {
        $stock_quantity = 30;
    } else {
        $stock_quantity = 10;
    }
    $sql3 = "INSERT INTO tbl_pds (pds_no,stock_name,quantity)
```

```
            VALUES ('$post_pds_no', '$stock_name', '$stock_quantity')";
                mysqli_query($conn, $sql3);
            }
            if (mysqli_query($conn, $sql)) {
                if (move_uploaded_file($tempname, $folder . $filename)) {
                    sleep(2);
                    echo '<script>
                      alert("Distributor Added Successfully...");
                      window.location.href="../admin/add_stock.php";
                      </script>';
                } else {
                    sleep(3);
                    echo '<script>alert("Distributor Not Added...")</script>';
                }
            } else
                echo "Error: " . $sql . "<br>" . mysqli_error($conn);
        }
    }
    ?>
<!DOCTYPE html>
<html lang="en" dir="ltr">
<head>
    <meta charset="UTF-8">
    <title> Admin | E-Ration </title>
    <link rel="stylesheet" href="style.css?v=<?= $v ?>">
    <link rel="stylesheet" href="https://www.w3schools.com/w3css/4/w3.css">
    <!-- Boxicons CDN Link -->
    <link href='https://unpkg.com/boxicons@2.0.7/css/boxicons.min.css' rel='stylesheet'>
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
</head>


<body>
    <div class="sidebar">
        <div class="logo-details">
            <img src="./images/logo2.png" height="40px" weight="40px" style="margin-left: 10px">
            <img src="./images/logo3.png" height="40px" weight="140px">
        </div>
        <ul class="nav-links">
            <li>
                <a href="admin_dash.php">
                    <i class='bx bx-grid-alt'></i>
                    <span class="links_name">Dashboard</span>
                </a>
            </li>
            <li>
                <a href="#" class="active">
                    <i class='bx bx-user'></i>
                    <span class="links_name">Add Distributor</span>
                </a>
            </li>
            <li>
                <a href="add_stock.php">
                    <i class="bx bxs-cart-add"></i>
                    <span class="links_name">Add Stock</span>
                </a>
            </li>
            <li>
                <a href="giveration.php">
```

```
        <i class="bx bx-coin-stack"></i>
        <span class="links_name">Give Stock</span>
      </a>
    </li>
    <li>
      <a href="applied_stock.php">
        <i class='bx bx-list-ul'></i>
        <span class="links_name">View Applied Stock</span>
      </a>
    </li>
    <li>
      <a href="stock_details.php">
        <i class='bx bx-coin-stack'></i>
        <span class="links_name">Stock Details</span>
      </a>
    </li>
    <li>
      <a href="stock_details.php">
        <i class='bx bx-user'></i>
        <span class="links_name">Distributors Details</span>
      </a>
    </li>
    <li>
      <a href="check_complaints.php">
        <i class='bx bx-message'></i>
        <span class="links_name">Check Complains</span>
      </a>
    </li>
    <li class="log_out">
```

```
                <a href="logout.php">
                    <i class='bx bx-log-out'></i>
                    <span class="links_name">Log out</span>
                </a>
            </li>
        </ul>
    </div>
    <section class="home-section">
        <nav>
            <div class="sidebar-button">
                <i class='bx bx-menu sidebarBtn'></i>
                <span class="dashboard">Add Distributor</span>
            </div>
            <div class="profile-details">
                <img src="images/profile.jpg" alt="">
                <span class="admin_name">Admin</span>
            </div>
        </nav>
        <div class="home-content">
            <div class="add-distributor">
                <div class="form_wrapper">
                    <div class="form_container">
                        <form method="post" enctype="multipart/form-data">
                            <div class="w3-row-padding">
                                <div class="w3-third w3-margin-top">
                                    <label>First Name</label>
                                    <input class="w3-input w3-border w3-margin-top"
name="first_name" type="text"
                                        placeholder="Enter First Name" required>
                                </div>
```

```
                <div class="w3-third w3-margin-top">
                  <label>Middle Name</label>
                  <input class="w3-input w3-border w3-margin-top"
name="middle_name" type="text"
                    placeholder="Enter Middle Name" required>
                </div>
                <div class="w3-third w3-margin-top">
                  <label>Last Name</label>
                  <input class="w3-input w3-border w3-margin-top"
name="last_name" type="text"
                    placeholder="Enter Last Name" required>
                </div>
              </div>
              <div class="w3-row-padding w3-margin-top">
                <div class="w3-third w3-margin-top">
                  <label>Email ID</label>
                  <input class="w3-input w3-border w3-margin-top" name="email"
type="email"
                    placeholder="Enter Email ID" required>
                </div>
                <div class="w3-third w3-margin-top">
                  <label>DOB</label>
                  <input class="w3-input w3-border w3-margin-top" name="dob"
type="date" required>
                </div>
                <div class="w3-third w3-margin-top">
                  <label>Phone Number</label>
                  <input class="w3-input w3-border w3-margin-top"
name="phone_no" type="tel"
                    placeholder="Enter Phone No." maxlength="10" required>
                </div>
```

```
                    </div>
                    <div class="w3-row-padding w3-margin-top">
                       <div class="w3-third w3-margin-top">
                          <label>Aadharcard No.</label>
                          <input class="w3-input w3-border w3-margin-top"
name="acard_no" type="number"
                             placeholder="Enter Aadharcard No." maxlength="12" required>
                       </div>
                       <div class="w3-third w3-margin-top">
                          <label>Rationcard No.</label>
                          <input class="w3-input w3-border w3-margin-top"
name="rcard_no" type="number"
                             placeholder="Enter Rationcard No." maxlength="15" required>
                       </div>
                       <div class="w3-third w3-margin-top">
                          <label>PDS No.</label>
                          <input class="w3-input w3-border w3-margin-top"
name="pds_no" type="number"
                             placeholder="Enter PDS No." required>
                       </div>
                    </div>
                    <div class="w3-row-padding w3-margin-top">
                       <div class="w3-third w3-margin-top">
                          <label>State</label>
                          <select class="w3-select w3-margin-top w3-border" name="state">
                             <option value="" disabled selected> Select your State
                             </option>
                             <option value="Tamilnadu">Tamilnadu</option>
                          </select>
                       </div>
                       <div class="w3-third w3-margin-top">
```

```
                    <label>City</label>
                    <select class="w3-select w3-margin-top w3-border" name="city">
                        <option value="" disabled selected> Select your City
                        </option>
                        <option value="Virudhunagar">Virudhunagar</option>
                        <option value="Sivakasi">Sivakasi</option>
                        <option value="Thiruthangal">Thiruthangal</option>
                        <option value="Rajapalayam">Rajapalayam</option>
                        <option value="Srivilliputhur">Srivilliputhur</option>
                    </select>
                </div>
                <div class="w3-third w3-margin-top">
                    <label>Pincode</label>
                    <input class="w3-input w3-border w3-margin-top"
name="pincode" type="num"
                        placeholder="Enter Pincode" maxlength="6" required>
                </div>
            </div>
            <div class="w3-row-padding w3-margin-top">
                <div class="w3-third w3-margin-top">
                    <label>Gender</label>
                    <div class="w3-padding-small">
                        <p>
                          <input class="w3-radio w3-margin-top w3-padding-small"
type="radio"
                                name="gender" value="male" checked>
                            <span>Male</span>
                        </p>
                        <p>
                    </div>
```

```
<div class="w3-padding-small">
    <input class="w3-radio w3-padding-small" type="radio"
name="gender"
        value="female">
    <span>Female</span></p>
</div>
<div class="w3-padding-small">
    <p>
        <input class="w3-radio w3-padding-small" type="radio"
name="gender"
            value="other">
        <span>Other</span>
    </p>
</div>
</div>
<div class="w3-third w3-margin-top">
    <label>Shop Address</label>
    <textarea rows="2" class="w3-margin-top w3-input w3-border"
        placeholder="Enter Address" name="address"
required></textarea>
</div>
<div class="w3-third w3-margin-top">
    <label>Password</label>                    <input class="w3-
input w3-border w3-margin-top" name="password" type="password"
        placeholder="Enter Password" required>
</div>
<div class="w3-third w3-margin-top">
    <label>Confirm Password</label>
    <input class="w3-input w3-border w3-margin-top"
name="con_password" type="password"
        placeholder="Enter Confirm Password" required>
```

```
                                </div>
                                <div class="w3-third w3-margin-top">
                                    <label>File Upload</label>
                                    <input type="file" id="myFile" name="uploadfile">
                                </div>
                            </div>
                    </div>
                    <div class="w3-center w3-margin-top">
                        <button class="w3-button w3-round-large w3-dark-blue" name="btn-add-
dist">SUBMIT</button>
                    </div>
                    </form>
                </div>
            </div>
        </div>
        </div>
    </section>
    <script>
    let sidebar = document.querySelector(".sidebar");
    let sidebarBtn = document.querySelector(".sidebarBtn");
    sidebarBtn.onclick = function() {
        sidebar.classList.toggle("active");
        if (sidebar.classList.contains("active")) {
            sidebarBtn.classList.replace("bx-menu", "bx-menu-alt-right");
        } else
            sidebarBtn.classList.replace("bx-menu-alt-right", "bx-menu");
    }
    </script>
</body>
</html>
```

```
<?php
} else {
    header("location: ../login/login.php");
}
?>
```

Login.php

```
<?php
    //This script will handle login
    session_start();
    if(isset($_SESSION['user']))
    {
        session_destroy();
    }
    /* check if the user is already logged in
    if(isset($_SESSION['username']))
    {
       header("location: distributor.php");
       exit;
    }*/
    include "connection.php";
    if(isset($_POST['btn-admin-login']))
    {
        $post_username=mysqli_real_escape_string($conn,$_POST['username']);
        $post_password=mysqli_real_escape_string($conn,$_POST['password']);
        if($post_username==$default_rc_no && $post_password==$default_password)
        {
            $_SESSION['user'] = $post_username;
            sleep(2);
            header("location: ../admin/admin_dash.php");
```

```
        }
        else
        {
            {
                sleep(2);
                echo '<script>alert("Please Enter Correct Crendentials of Admin")</script>';
            }
        }
    }
?>
<!DOCTYPE html>
<html lang="zxx">
<head>
            <title>Login Page</title>
            <!-- Meta tag Keywords -->
            <meta name="viewport" content="width=device-width, initial-scale=1">
            <meta charset="UTF-8" />
            <meta name="keywords"       content="Working Signin form Responsive web
template, Bootstrap Web Templates, Flat Web Templates, Android Compatible web
template, Smartphone Compatible web template, free webdesigns for Nokia, Samsung,
LG, SonyEricsson, Motorola web design" />
            <link rel="stylesheet" href="https://www.w3schools.com/w3css/4/w3.css">
<link rel="stylesheet" href="https://www.w3schools.com/lib/w3-theme-blue-grey.css">
    <!-- //Meta tag Keywords -->
            <link
href="//fonts.googleapis.com/css2?family=Karla:wght@400;700&display=swap"
rel="stylesheet">
            <!--/Style-CSS -->
            <link rel="stylesheet" href="style.css" type="text/css" media="all" />
            <!--//Style-CSS -->
</head>
```

```
<body>
            <!-- form section start -->
            <section class="w3l-workinghny-form " >
        <!-- /form -->
        <div class="workinghny-form-grid">
            <div class="wrapper">
                <div class="logo">
                    <h1><a class="brand-logo" style="pointer-events:none;"> Log In  </a></h1>
                    <!-- if logo is image enable this
                        <a class="brand-logo" href="#index.html">
                            <img src="image-path" alt="Your logo" title="Your logo"
style="height:35px;" />
                        </a> -->
                </div>
                <div class="workinghny-block-grid">
                    <div class="workinghny-left-img align-end">
                        <img src="2.png" class="img-responsive" alt="img" />
                    </div>
                    <div class="form-right-inf">
                        <div class="login-form-content">
                            <form class="signin-form" method="post">
                                                                        <div class="one-frm">

            <label>Username</label>
            <input type="text" name="username" id="username" placeholder="Enter
Username" required="">
                                                                        </div>
                                                                <div class="one-frm">

            <label>Password</label>
```

```
                                                            <input type="password"
name="password" id="password" placeholder="Enter Password" required="">
                                                </div>

                                <button type="submit" class="btn btn-style mt-3 w3-theme-d2"
name="btn-admin-login">Sign In </button>

                                <p class="already">Are You Customer ? <a href="login.php">Login
Here</a></p>

                            </form>

                        </div>

                    </div>

                </div>

            </div>

        </div>

        <!-- //form -->

    </section>

    <!-- //form section start -->

</body>

</html>
```

Connection.php

```
<?php

$servername = "localhost";

$db_username = "root";

$db_password = "";

$db= "kumar";

$default_rc_no="admin";

$default_password="admin";

// Create connection

$conn = mysqli_connect($servername, $db_username, $db_password, $db);

// Check connection

if (!$conn) {

  die("Connection failed: " . mysqli_connect_error());
```

```
  echo "<br>";
}
?>
```

Logout.php

```
<?php
session_start();


if(!isset($_SESSION['rationcard_no']))
{
	header("location: ../Auth/login.php");
}
else
{
	session_destroy();
	header("location: ../Auth/login.php");
}
```

Script.js

```
let sidebar = document.querySelector(".sidebar");
let sidebarBtn = document.querySelector(".sidebarBtn");
sidebarBtn.onclick = function() {
  sidebar.classList.toggle("active");
  if(sidebar.classList.contains("active")){
  sidebarBtn.classList.replace("bx-menu" ,"bx-menu-alt-right");
}else
  sidebarBtn.classList.replace("bx-menu-alt-right", "bx-menu");
}
```

Update_cart.php

```
<?php
session_start();
```

```
include 'connection.php';

$rcard_no = $_SESSION['rationcard_no'];

$cart = $_SESSION['count'];

if (isset($cart)) {

    echo "<script>alert('If You logged out , Your Product has been Removed from Cart..!')</script>";

    echo "<script>window.location='../Auth/login.php'</script>";

} else {

    echo "<script>alert('Product has been not Removed..!)</script>";

}
```

Test.php

```
<?php

include "connection.php";

$sql = "SELECT count(*) as count1 FROM tbl_user";


$run = mysqli_query($conn, $sql);


while($row = mysqli_fetch_array($run))

{

  print_r(count1);

}

?>
```

Autoload.php

```
<?php

// autoload.php @generated by Composer

if (PHP_VERSION_ID < 50600) {

    if (!headers_sent()) {

        header('HTTP/1.1 500 Internal Server Error');

    }
```

```
    $err = 'Composer 2.3.0 dropped support for autoloading on PHP <5.6 and you are running '.PHP_VERSION.', please upgrade PHP or use Composer 2.2 LTS via "composer self-update --2.2". Aborting.'.PHP_EOL;
    if (!ini_get('display_errors')) {
        if (PHP_SAPI === 'cli' || PHP_SAPI === 'phpdbg') {
            fwrite(STDERR, $err);
        } elseif (!headers_sent()) {
            echo $err;
        }
    }
    throw new RuntimeException($err);
}
require_once __DIR__ . '/composer/autoload_real.php';
return ComposerAutoloaderInit19dce85914a0d40ff610e42a7553729f::getLoader();
```

Deprecated.php

```
<?php
/**
 * Backwards compatibility layer for Requests.
 *
 * Allows for Composer to autoload the old PSR-0 classes via the custom autoloader.
 * This prevents issues with _extending final classes_ (which was the previous solution).
 *
 * Please see the Changelog for the 2.0.4 release for upgrade notes.
 *
 * @package Requests
 *
 * @deprecated 2.0.4 Use the PSR-4 class names instead.
 */
define("REQUESTS_SILENCE_PSR0_DEPRECATIONS",true);
```

```
if (class_exists('WpOrg\Requests\Autoload') === false) {
	require_once __DIR__. 'libs/Requests-2.0.4/src/Autoload.php';
}
WpOrg\Requests\Autoload::register();
```

Autoload_classmap.php

```
<?php
// autoload_classmap.php @generated by Composer
$vendorDir = dirname(__DIR__);
$baseDir = dirname($vendorDir);
return array(
    'Composer\\InstalledVersions' => $vendorDir . '/composer/InstalledVersions.php',
    'Requests' => $vendorDir . '/rmccue/requests/library/Requests.php',
);
```

Autoload_files.php

```
<?php
// autoload_files.php @generated by Composer
$vendorDir = dirname(__DIR__);
$baseDir = dirname($vendorDir);
return array(
    '941748b3c8cae4466c827dfb5ca9602a' => $vendorDir .
'/rmccue/requests/library/Deprecated.php',
    '13906c19e3d8fcd1341b24ed4d51cf72' => $vendorDir .
'/razorpay/razorpay/Deprecated.php',
);
```

Autoload_namespaces.php

```
<?php
// autoload_namespaces.php @generated by Composer
$vendorDir = dirname(__DIR__);
$baseDir = dirname($vendorDir);
return array();
```

Autoload_psr4.php

```
<?php
// autoload_psr4.php @generated by Composer
$vendorDir = dirname(__DIR__);
$baseDir = dirname($vendorDir);
return array(
    'WpOrg\\Requests\\' => array($vendorDir . '/rmccue/requests/src'),
    'Razorpay\\Tests\\' => array($vendorDir . '/razorpay/razorpay/tests'),
    'Razorpay\\Api\\' => array($vendorDir . '/razorpay/razorpay/src'),
    'PHPMailer\\PHPMailer\\' => array($vendorDir . '/phpmailer/phpmailer/src'),
);
```

Platform_check.php

```
<?php
// platform_check.php @generated by Composer
$issues = array();
if (!(PHP_VERSION_ID >= 70300)) {
    $issues[] = 'Your Composer dependencies require a PHP version ">= 7.3.0". You are running ' . PHP_VERSION . '.';
}
if ($issues) {
    if (!headers_sent()) {
        header('HTTP/1.1 500 Internal Server Error');
    }
   if (!ini_get('display_errors')) {
        if (PHP_SAPI === 'cli' || PHP_SAPI === 'phpdbg') {
            fwrite(STDERR, 'Composer detected issues in your platform:' . PHP_EOL.PHP_EOL . implode(PHP_EOL, $issues) . PHP_EOL.PHP_EOL);
        } elseif (!headers_sent()) {
            echo 'Composer detected issues in your platform:' . PHP_EOL.PHP_EOL . str_replace('You are running '.PHP_VERSION.'.', '', implode(PHP_EOL, $issues)) . PHP_EOL.PHP_EOL;
```

```
        }
    }
    trigger_error(
        'Composer detected issues in your platform: ' . implode(' ', $issues),
        E_USER_ERROR
    );
}
```

Print_customers.php

```
<?php
    session_start();
    if(isset($_SESSION['rationcard_no']))
    {
    $rcard_no=$_SESSION['rationcard_no'];
    $d_pincode=$_REQUEST['pincode'];
    include ('../php/connection.php');
    $sql1="SELECT * FROM tbl_user WHERE pincode='$d_pincode'";
    $result1=mysqli_query($conn,$sql1);
    $d_customers=mysqli_fetch_all($result1,MYSQLI_ASSOC);
    $sql2="SELECT * FROM tbl_distributor WHERE rationcard_no='$rcard_no'";
    $result2=mysqli_query($conn,$sql2);
    $d_pds=mysqli_fetch_assoc($result2);
    $n=1;
        $pds=$d_pds['pds_no'];
    ?>
<!DOCTYPE html>
  <html lang="en" dir="ltr">
```

```
<head>
    <meta charset="UTF-8">
    <title> Distributor | E-Ration </title>
    <link href="https://fonts.googleapis.com/icon?family=Material+Icons"
rel="stylesheet">
    <link rel="stylesheet" href="https://www.w3schools.com/w3css/4/w3.css">
    <style>
    .table{
    width: 50%;
    border-collapse: collapse;
    margin-left: auto;
    margin-right: auto;
    }
    .table thead{
    background-color:#0A2558;
    }
    table thead tr th{
    font-size: 14px;
    font-weight: 600;
    letter-spacing: 0.35px;
    color: #ffffff;
    opacity: 1;
    padding: 12px;
    vertical-align: top;
    border: 1px solid #dee2e685;
    }
```

```
.table tbody tr td{

font-size: 14px;

font-weight: normal;

letter-spacing: 0.35px;

padding: 8px;

text-align: center;

border: 1px solid #dee2e685;

}

</style>

</head>

<body>

<div class="w3-container w3-center">

<h2 style="margin-top: 50px;">E-Ration</h2>

<h4 style="margin-top: -10px;">Customer Details Under PDS NO-<?php
echo $pds; ?></h4>

<h5>Date:<?php

date_default_timezone_set("Asia/Kolkata");

echo date("d-m-
y")."            


            "."
Time: ".date("h:i:s a"); ?></h5>

</div>

<table class="table">

<thead>

<tr>
```

```
            <th>SR No</th>
            <th>Name</th>
            <th>Ration Card No</th>
            <th>Aadhar Card No</th>
            <th>DOB</th>
            <th>Email-id</th>
            <th>Phone No</th>
            <th>Pincode</th>
            <th>Address</th>
        </tr>
        </thead>
        <tbody>
        <?php
            foreach($d_customers as $d_customer)
            {
            ?>
            <tr>
                <td data-label="SR No"><?php echo $n; ?></td>
                <td data-label="Name"><?php echo $d_customer['fname']."
".$d_customer['mname']." ".$d_customer['lname']; ?></td>
                <td data-label="Ration Card No"><?php echo $d_customer['rationcard_no'];
?></td>
                <td data-label="Aadhar Card No"><?php echo $d_customer['aadhar_no'];
?></td>
                <td data-label="DOB"><?php echo $d_customer['dob']; ?></td>
                <td data-label="Email-id"><?php echo $d_customer['email_id']; ?></td>
                <td data-label="Phone No"><?php echo $d_customer['contact_no']; ?></td>
```

```
                <td data-label="Pincode"><?php echo $d_customer['pincode']; ?></td>

                <td data-label="Address"><?php echo $d_customer['address']; ?></td>

            </tr>

            <?php

                $n++;

            }

        ?>

        </tbody>

    </table>

    <button type="submit" onclick="myFunction()" id="printpagebutton" style="font-
size:24px;margin-left: 47%;

    margin-top:10px;margin-bottom: 10px;background-color: #0A2558;color:
white;">Print <i class="material-icons"></i></button>

                <script>

                        function myFunction() {

                                var printButton =
document.getElementById("printpagebutton");

                                        printButton.style.visibility = 'hidden';

                                        window.print()

                                        printButton.style.visibility = 'visible';

                        }

                </script>

  </body>

 </html>

 <?php

}

else{
```

```
        header("location: ../login/login.php");
}
?>
```

Success.php

```
<?php
    include 'connection.php';
    $tid = $_POST['txn_id'];
    $penalty = $_POST['payment_gross'];
    $price=$penalty*75;
    $dollar=$price/75;
    if($penalty == $dollar)
    {
        $int=$price;
    }
    $state = $_POST['payment_status'];
    $payer_id = $_POST['payer_id'];
    $rcard_no = $_POST['item_name'];
    $u_id = $_POST['item_number'];
    date_default_timezone_set("Asia/Calcutta");
    $date = date("Y-m-d");
    $time = date("H:i:s");
```

```
    //fetching bookin_id
  $sql1="SELECT * FROM tbl_book WHERE rationcard_no='$rcard_no' AND
date='$date'";
  $result1=mysqli_query($conn,$sql1);
  $booking=mysqli_fetch_assoc($result1);
  $booking_id=$booking['booking_id'];
  //inserting into payment table with refernece to booking_id
  $sql2="INSERT INTO tbl_payment (mode, date, time, amount, booking_id)
  VALUES ('Online', '$date', '$time', '$int', '$booking_id')";
  mysqli_query($conn,$sql2);
  //fetching distributor id with pincode and customer pincode
  $sql3="SELECT * FROM tbl_user WHERE u_id='$u_id'";
  $result3=mysqli_query($conn,$sql3);
  $user_info=mysqli_fetch_assoc($result3);
  $u_pincode=$user_info['pincode'];
  $u_name=$user_info['fname']." ". $user_info['mname']." ". $user_info['lname'];
  $u_ph=$user_info['contact_no'];
  $sql4="SELECT * FROM tbl_distributor WHERE pincode='$u_pincode'";
  $result4=mysqli_query($conn,$sql4);
  $dist_info=mysqli_fetch_assoc($result4);
  $d_id=$dist_info['d_id'];
```

```
//inserting into receipt table
    $sql5="INSERT INTO tbl_receipt (u_name, u_contact_no, amount, rationcard_no,
date, time, tid, state, d_id, booking_id)
    VALUES ('$u_name', '$u_ph', '$int', '$rcard_no', '$date', '$time', '$tid', '$state', '$d_id',
'$booking_id')";
    if(mysqli_query($conn,$sql5))
    {
        //fetching receipt_id from receipt table
        $sql6="SELECT * FROM tbl_receipt WHERE tid='$tid'";
        $result6=mysqli_query($conn,$sql6);
        $r_info=mysqli_fetch_assoc($result6);
        $r_id=$r_info['receipt_id'];
 //updating booking status
        $sql7="UPDATE tbl_book SET status=1 WHERE booking_id='$booking_id'";
        if(mysqli_query($conn,$sql7))
        {
            //here comes header function which redirects to receipt printing page
            header("Location:
receipt.php?tid=$tid&total_amount=$int&rationcard_no=$rcard_no");
        }
        else
```

```
    {
        echo '<script>alert("Something Went Wrong")</script>';
    }
}
?>
```

REFERENCES

1. https://www.php.net/manual/en/

2. https://www.w3schools.com/php/

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